# Lecture Notes

Mathematical Economics Class

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## 1 Introduction

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The courses is designed to facilitate a smooth transition into a first year PhD program in Economics by providing the the basic mathematical foundation needed by incoming Economic PhD students. By focusing on intuitive understanding and practical applications, the classes are taught as a complement to a background course in Mathematics, such as the one held by Prof. Villanacci at the EUI. Over the five classes, if time allows, we will try to cover the following concepts:

1. Class 1 - Thursday: Multivariate calculus:
   (a) The Chain Rule
   (b) The Total Derivative
   (c) The Implicit Function Theorem
2. Class 2 - Friday: Optimization:
   (a) Unconstrained optimization
   (b) Introduction to Lagrangian approach
   (c) Introduction to Kuhn-Tucker - Part I
3. Class 3: Monday - Dynamic optimization:
   (a) Introduction to Kuhn-Tucker - Part II
   (b) Introduction to Dynamic programming
4. Class 4: Tuesday - Matrices:
   (a) Fundamentals of Matrix algebra
   (b) The determinant of a matrix
   (c) The inverse of a matrix
   (d) Eigenvalues and Eigenvectors

5. Class 5: Wednesday - Advanced Matrix Topics:
   (a) Gradient vs. Jacobian
   (b) Computing the Hessian matrix
   (c) Analysis of Matrix Definiteness (Positive Definite, Negative Definite, Positive Semidefinite, and Negative Semidefinite Matrices)

The main reference are the lecture notes based on:

- Simon, C.P. and Blume, L., 1994. Mathematics for economists (Vol. 7). New York: Norton.
- Villanacci Antonio. Lecture Notes. Background Course on Mathematics.
- Cooper Russell. Lecture Notes. Overview of Applied Dynamic Programming.

Other useful references:

- Arizona University Economic Department Math Camp (fully online). Link: `https://www.u.arizona.edu/~mwalker/MathCamp2021.htm`
- Geeksforgeeks.org (very useful to better understand specific concepts). Link: `https://www.geeksforgeeks.org/`

# 2 Lecture 1

In this lecture, we will see some useful tools to study functions.

## 2.1 The Chain Rule

Given two functions $g(x)$ and $h(x)$ defined from $\mathbb{R}$ to $\mathbb{R}$, consider the function formed by first applying $g$ to any number $x$ and then applying function $h$ to the result $g(X)$. Such a function is called **composite function** and can be denoted as $f(x) = h(g(x))$ or as $f(x) = (h \circ g)(x)$.

$$x \longmapsto g(x) \longmapsto \underset{\substack{\equiv \\ f(x)}}{h(g(x))}$$

*Example*: The function $f(x) = e^x + 1$ can be seen as $f(x) = h(g(x))$ where $g(x) = e^x$ and $h(x) = x + 1$. □

We are interested in studying the derivative of a composite function and leveraging the fact that is composed of multiple functions. For this purpose, we use the Chain Rule.

The **Chain Rule** states that for a function $f(x) = h(g(x))$ the derivative is

$$\frac{df}{dx}(x) = \frac{d}{dx}(h(g(x)) = h'(g(x)) \cdot g'(x)$$

In words, the derivative of a composite function can be thought of as the derivative of the *outside* function ($h$) times the derivative of the *inside* function ($g$). Be aware that to apply the Chain Rule the function $h$ and $g$ have to be differentiable. [1]

For clarity and to get used to the notation, the Chain Rule can also be written as

$$\frac{df}{dx}(x) = \frac{d(h \circ g)}{dx}(x) = \frac{dh}{dz}(g(x)) \cdot \frac{dg}{dx}(x)$$

where z is the argument of the outside function ($h(z)$), or, in the Mas-Colell, as

$$D_x h(g(x)) = Dh(g(x))Dg(x)$$

where $D\cdot$ is the Jacobian matrix and will be discussed in the last Lecture.

*Example*: Let's differentiate the function $U(x) = (\alpha x^\rho + 1)^{\frac{1}{\rho}}$ with $\alpha$ and $\rho$ as parameters, this function can be seen as $U(x) = h(g(x))$ where $g(x) = (\alpha x^\rho + 1)$ is the inside function and $h(x) = x^{\frac{1}{\rho}}$ as outside function. The derivative of the outside function evaluated at the inside function is $h'(g(x)) = \frac{1}{\rho}(g(x))^{\frac{1}{\rho}-1} = \frac{1}{\rho}(g(x))^{\frac{1-\rho}{\rho}} = \frac{1}{\rho}(\alpha x^\rho + 1)^{\frac{1-\rho}{\rho}}$, while the derivative of the inside function is $g'(x) = (\alpha\rho x^{\rho-1})$. By applying the Chain Rule I get:

$$\frac{df}{dx}(x) = \frac{d(h \circ g)}{dx}(x) = \frac{dh}{dz}(g(x)) \cdot \frac{dg}{dx}(x) = \frac{1}{\rho}(\alpha x^\rho + 1)^{\frac{1-\rho}{\rho}} \cdot (\alpha\rho x^{\rho-1}) = \alpha x^{\rho-1}(\alpha x^\rho + 1)^{\frac{1-\rho}{\rho}}$$

□

Similarly, the rule can be applied if a function is composed of more than two functions.

*Example*: Let's derive the function $f(x) = [log(x^{\frac{1}{3}})]^2$

$$\frac{df}{dx}(x) = 2[log(x^{\frac{1}{3}})]^{2-1} \cdot \frac{1}{x^{\frac{1}{3}}} \cdot \frac{1}{3}x^{\frac{1}{3}-1} = \frac{2}{3}\frac{1}{x}log(x^{\frac{1}{3}})$$

□

What happens if we have two inside functions? if $f(x) = h(g_1(x), g_2(x))$? We apply the Chain Rule for each component and sum the results:

$$\frac{\partial f}{\partial x}(x) = \frac{dh}{dx}(g_1(x), g_2(x) = \frac{\partial h}{\partial g_1}(g_1(x), g_2(x)) \cdot \frac{dg_1}{dx}(x) + \frac{\partial h}{\partial g_2}(g_1(x), g_2(x)) \cdot \frac{dg_2}{dx}(x)$$

or more compactly, as mentioned before, as

$$D_x h(g(x)) = Dh(g(x))Dg(x)$$

*Example*: Let's differentiate the function $V(k) = u(f(k) - k)$

$$\frac{\partial V}{\partial k}(k) = u'(f(k) - k) \cdot f'(k) + u'(f(k) - k) \cdot (-1) = u'(f(k) - k) \cdot (f'(k) - 1)$$

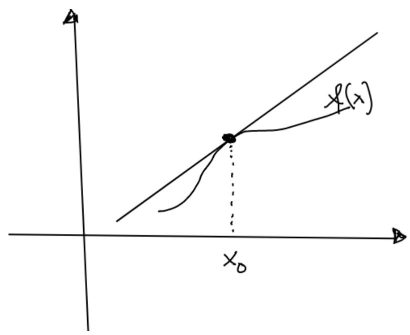
[1] If $g$ and $h$ are both differentiable functions, then the composite function $f(x) = h(g(x))$ is also differentiable.

Note that in the above differentiation we are considering $\frac{\partial V}{\partial k}(x) = \frac{dh}{dk}(g_1(k), g_2(k))$ where $g_1(k) = f(k)$ and $g_2(k) = -k$. Alternatively we could have simply considered as outside function $u()$ and as inside function all $f(k) - k$, and applying directly the simple version of the chain rule.

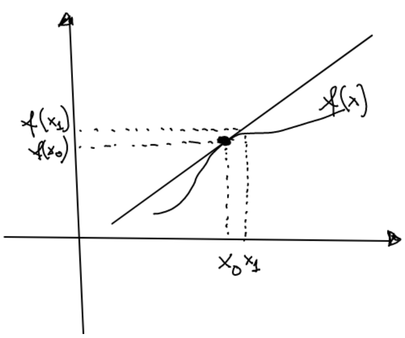
□

## 2.2 The Total Derivative

From your previous studies, you should be aware that in $R^2$ is possible to approximate the value of a function in a point $x_0$ by making use of the derivative:




More specifically, the value of the function $f(x)$ in $x_0$ can be approximated by the line $f(x_0) + f(x_0)'(x - x_0)$.




Here $f(x_1) \approx f(x_0) + f(x_0)'(x_1 - x_0)$. It is possible to see that the derivative can be used as a linear approximation of a function near a point: $f(x_1) - f(x_0) \approx f(x_0) + f(x_0)'\Delta x$.

The same can happen when we are in a multivariate world. The value of the function $f(x_1, ..., x_n)$ around $(x_{1,0}, ..., x_{n,0})$ can be approximated by

$$f(x_1, ..., x_n) \approx f(x_{1,0}, ..., x_{n,0}) + \frac{\partial f}{\partial x_1}(x_{1,0}, ..., x_{n,0})\Delta x_1 + \frac{\partial f}{\partial x_2}(x_{1,0}, ..., x_{n,0})\Delta x_2 + ... + \frac{\partial f}{\partial x_n}(x_{1,0}, ..., x_{n,0})\Delta x_n$$

Note that if in $\mathbb{R}^2$ we were graphically interpreting the approximation as a line tangent to the curve, in $\mathbb{R}^3$ we are approximating the function with the plane tangent to the curve.

If one is interested in understanding how the value of a function changes when all its input variables change, one should compute:

$$f(x_1, ..., x_n) - f(x_{1,0}, ..., x_{n,0}) \approx \frac{\partial f}{\partial x_1}(x_{1,0}, ..., x_{n,0})\Delta x_1 + \frac{\partial f}{\partial x_2}(x_{1,0}, ..., x_{n,0})\Delta x_2 + ... + \frac{\partial f}{\partial x_n}(x_{1,0}, ..., x_{n,0})\Delta x_n$$

as the change becomes infinitesimal ($\Delta \to \infty$), we get

$$df \approx \frac{\partial f}{\partial x_1}(x_{1,0},...,x_{n,0})dx_1 + \frac{\partial f}{\partial x_2}(x_{1,0},...,x_{n,0})dx_2 + ... + \frac{\partial f}{\partial x_n}(x_{1,0},...,x_{n,0})dx_n$$

the RHS of the equation is called the total differential of a function and is denoted as $df$. Note that the elements in the RHS are the multiplication of partial derivatives with the marginal changes. If we consider the gradient (or row vector) evaluated at $(x_{1,0},...,x_{n,0})$, that is the vector of partial derivatives evaluated in that point:

$$\nabla f(x_{1,0},...,x_{n,0}) = \begin{pmatrix} \frac{\partial f}{\partial x_1}(x_{1,0},...,x_{n,0}) \\ \frac{\partial f}{\partial x_2}(x_{1,0},...,x_{n,0}) \\ \vdots \\ \frac{\partial f}{\partial x_n}(x_{1,0},...,x_{n,0}) \end{pmatrix}$$

and the vector of marginal changes

$$\nabla dx = \begin{pmatrix} dx_1 \\ dx_2 \\ \vdots \\ dx_n \end{pmatrix}$$

we can rewrite the total differential in more compact form.

In other words, the **total differential** is

$$df = \nabla f(x_{1,0},...,x_{n,0}) \cdot dx$$

where the $\nabla f(x_{1,0},...,x_{n,0})$ is the gradient of the function and $dx$ is the vector of marginal changes.

*Example*: Let's found a linear approximation that describes what happens to the function $F(x_1,x_2) = x_1^{\frac{2}{3}}x_2^{\frac{1}{3}}$ when we slightly change $x_1$ and $x_2$, that is we compute the total derivative:

$$\begin{aligned} dF &= \frac{\partial x_1^{\frac{2}{3}}x_2^{\frac{1}{3}}}{\partial x_1}dx_1 + \frac{\partial x_1^{\frac{2}{3}}x_2^{\frac{1}{3}}}{\partial x_2}dx_2 \\ &= \frac{2}{3}x_1^{-\frac{1}{3}}x_2^{\frac{1}{3}}dx_1 + \frac{1}{3}x_1^{\frac{2}{3}}x_2^{-\frac{2}{3}}dx_2 \end{aligned}$$

□

The total derivative can also be used to study an optimal point.

*Example*: Consider the problem from Macroeconomics II lecture

$$\max_n u(wn + A) - g(n)$$

where

- $u$ is the utility function with $u'() > 0$, $u''() < 0$ (strictly increasing and concave)

- g is the disutility function with $g'() > 0$, $g'' > ()0$ (strictly increasing and convex)
- $u'(0) > g'(0)$ and $u'(1) < g'(1)$
- $w$ is the salary, $0 \leq n \leq 1$ is the amount of hours worked and $A$ is a fix income

We want to find the optimal amount of hours worked and determine how this is influenced by changes in $A$.
To find the optimal point, we take the derivative of the function w.r.t. $n$ and set it equal to zero. The first order condition (F.O.C.) is

$$\begin{aligned} wu'(wn + A) - g'(n) &= 0 \\ wu'(wn + A) &= g'(n) \end{aligned}$$

The optimal value of working hours must satisfy the above equation. In other words, every value of $n$ that satisfies the above equation is an optimal value of $n$ ($n^*$). To make this more evident we could write the above equation as

$$wu'(wn^* + A) = g'(n^*)$$

but to simplify the notation, we will simply use $n$.

We are now interested in understanding what happens to the optimal value once we change the value of $w$. To do so, we will use the total derivative. We want to apply the total derivative to the RHS and to the LHS, so we will have the total derivative of $F_1 = wu'(wn + A)$ and the total derivative of $F_2 = -g'(n)$, and near the optimal value of $n$ the two total derivative must be equal:

$$dF_1 = dF_2$$

We compute them

$$\begin{aligned} dF_1 &= [w^2u''()]dn + [u'() + wnu''()]dw + [wu''()]dA \\ dF_2 &= [g''()]dn \end{aligned}$$

We put them equal and get

$$[w^2u''()]dn + [u'() + wnu''()]dw + [wu''()]dA = [g''()]dn$$

which can be rearranged as

$$-[w^2u''() - g''()]dn = [u'() + wnu''()]dw + [wu''()]dA$$

Since we are only interested in $\frac{dn}{dA}$ (how does $n$ changes when we change $A$) while $w$ stays the same, we can set $dw = 0$ and get

$$\frac{dn}{dA} = -\frac{wu''()}{[w^2u''() - g''()]} < 0$$

where the value is smaller than zero because the numerator is negative by assumption and the denominator is positive (following the assumptions). Therefore we can say that when $A$ increases, the optimal value of $n$ decreases, which makes sense since when the fix income is increasing, we are decreasing the optimal amount of hours worked. □

## 2.3 The Implicit Function Theorem

### 2.3.1 Intuition

In economics, we often have to deal with equations that include both exogenous and endogenous variables. These equations can be expressed in two main forms:

- A function is written in **explicit form** when we have

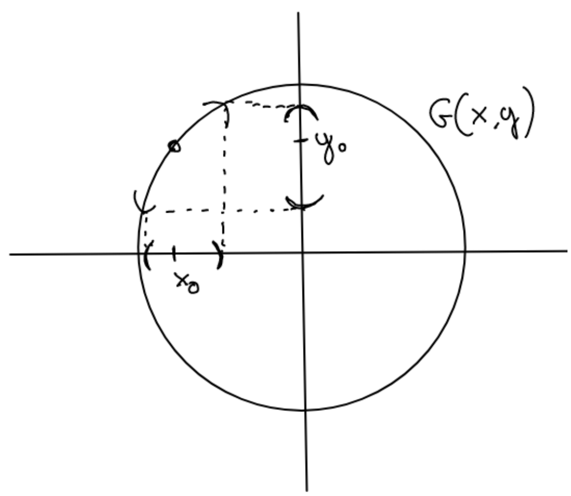
$$y = F(x) \text{ in } R^2 \text{ or } y = F(x_1, ..., x_n) \text{ in } R^{n+1}$$

  so that the endogenous variable $y$ is an explicit function of the exogenous variables $x_1, ..., x_n$;

- An alternative is to write the equation in **implicit form**

$$G(x, y) = 0 \text{ in } R^2 \text{ or } G(x_1, ...., x_n, y) = 0 \text{ in } R^{n+1} \tag{1}$$

*Example*: Consider the formula of the circle of radius one and centered at the origin. Its implicit functional form is $x^2 + y^2 = 1$ (or $x^2 + y^2 - 1 = 0$) and the explicit form is $y = \pm(1 - x^2)^{\frac{1}{2}}$.




□

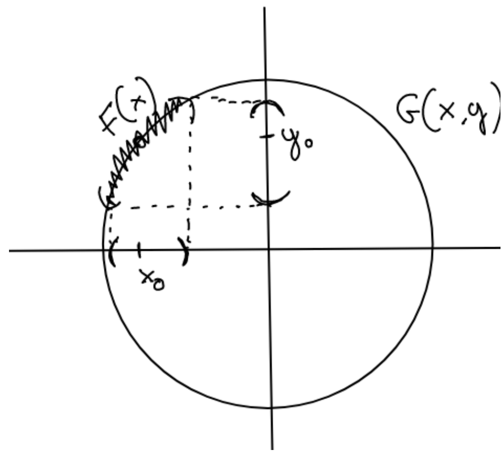
In $\mathbb{R}^2$, suppose we would like to know

1. **Existence of a function:** if, given the implicit functional form and a point $(x_0, y_0)$ such that $G(x_0, y_0) = 0$, a continuous function describing how $y$ behave as a function of $x$ *exists* close to $(x_0)$

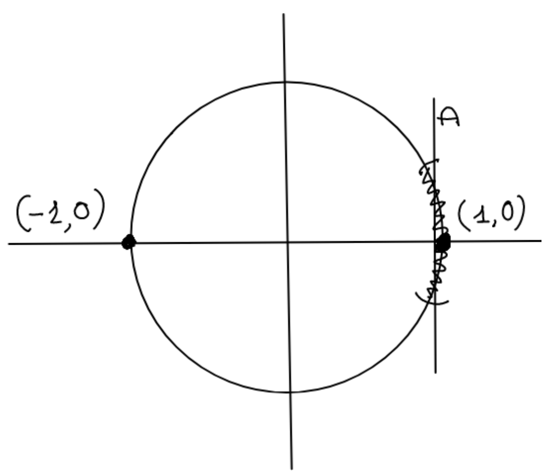
2. **Rate of Change:** how does the $y$ *changes* when we change the x value a bit?

Which tool can we use to answer the two questions above? The Implicit Function Theorem (IFT). It tells us that, under some assumptions, a specific function exists, and it allows us to compute its derivative directly from the implicit function.

*Example:* Let's go back to the example of the circle $G(x, y) = x^2 + y^2 - 1 = 0$ and we consider a random point on it. I want to see if, in an interval around the point, I can rewrite $G(x, y)$ in explicit form, that is as a function that goes from an interval around $x_0$ to an interval around $y_0$. I will look for a function F(x):




So we start with a specific point $(x_0, y_0)$ where $G(x, y) = 0$ hold, for example $(0, 1)$ and we vary $x$ a little and we can find a unique $y = \sqrt{1-x^2}$ near $y = 1$. With the circle, this can always be done except in (1,0) and (-1,0). In a because in an interval of those points no function from $x$ to $y$ can be defined. Why? Because for points with $x$ greater than 1, no function can be defined. and in point smaller than one. For points with $x$ smaller than one, you have two equally good candidates for $y$ near $y = 0$: $y = \sqrt{1-x^2}$ and $y = -\sqrt{1-x^2}$. So the difference is that before, given a point you were uniquely choosing a specific functional form, now given a point you are not sure which of the two to use. (1,0) and (-1,0) corresponds to locations where $\frac{\partial G(x,y)}{\partial y} = 0$.




□

### 2.3.2 Formal Definition

For the IFT to hold the following assumptions needs to hold

- A function $G(x, y) : A \to \mathbb{R}$, where $A$ is open
- The function and its partial derivatives are continuous in A
- At a point $(x_0, y_0) \in A$ it must be that
  - $G(x_0, y_0) = 0$
  - $\frac{\partial G(x_0,y_0)}{\partial y} \neq 0$

Note that the partial derivative is computed with respect to the endogenous variable.[2]

*Example*: In the example of the circle with $A = (0, 1)$, all the assumptions are satisfied. Note that the points $(1, 0)$ and $(-1, 0)$ are excluded from the domain; here, we can not apply the IFT because the third assumption is violated. As seen above, no function from $x$ to $y$ can be defined here. $\square$

According to the **Implicit Function Theorem**, then it exists a unique *continuous* function $F = F(x)$ defined on an interval $I$ around $x_0$ such that we have

- $G(x, F(x)) = 0$ for all $x$ in $I$
- $F(x_0) = y_0$
- If the partial derivatives of $G$ are continuous, then $F$ is derivable and

$$F'(x) = -\frac{\frac{\partial G}{\partial x}(x, y)}{\frac{\partial G}{\partial y}(x, y)}$$

  Note that this formula holds for all $x \in I$, in particular at $x = x_0$

The IFT provides the conditions to determine the existence of a function (first two points) and the rate of change (third point). During the year, you will especially make use of the latter one since it allows to easily study how does $y$ changes when one changes the value of $x$.

***IFT Recipe***

1. Write down the function as $G(x, y)$ and compute the derivative of the endogenous variable ($\frac{\partial G(x,y)}{\partial y}$)
2. Identify the exogenous (x) and the endogenous (y) variables.
3. Check if the assumption holds
   - Check if the domain is open

[2] To help you connect to the first block Math course, be aware that in the $\mathbb{R}^n$ version of the Theorem, this condition corresponds to the rank condition, which, as you will see, implies that the determinant of the partial Jacobian of the endogenous variables is different from zero.

- Check if the function and its derivatives are continuous in the domain
- Select the points where $G(x, y) = 0$ and the partial derivative is different from zero ($\frac{\partial G(x_0,y_0)}{\partial y} \neq 0$)

4. Apply the theorem and compute the derivative

$$F'(x) = -\frac{\frac{\partial G}{\partial x}(x, y)}{\frac{\partial G}{\partial y}(x, y)}$$

*Example:* Once again, we focus on the circle

1. We write down the function and compute the derivative

$$\begin{aligned} G(x, y) &= x^2 + y^2 - 1 \\ \frac{\partial G(x, y)}{\partial y} &= 2y \\ \frac{\partial G(x, y)}{\partial x} &= 2x \end{aligned}$$

2. $x$ is the exogenous variable and $y$ the endogenous one
3. We check the assumptions
   - The function and the derivatives are continuous in the domain
   - We select as points $(0, 0), (0, 1), (1, 0)$
     - $(0, 0)$: $G(0, 0) = 0 + 0 - 1 = -1$ which violates the assumption of $G(x_0, y_0) = 0$, therefore the IFT cannot be applied at $(0, 0)$
     - $(0, 1)$: $G(0, 1) = 0 + 1 - 1 = 0$ and $\frac{\partial G(0,1)}{\partial y} = 2$
     - $(1, 0)$: $G(1, 0) = 1+0-1 = 0$ and $\frac{\partial G(1,0)}{\partial y} = 0$, therefore the IFT cannot be applied at $(1, 0)$ by solving for $y$.
4. We apply the IFT at $(0, 1)$ and compute the derivative

$$F'(x) = -\frac{\frac{\partial G}{\partial x}(0, 1)}{\frac{\partial G}{\partial y}(0, 1)} = -\frac{2 \cdot 0}{2 \cdot 1} = -\frac{0}{2} = 0$$

□

*Example*: Apply the IFT to study the example from Macroeconomics II Lectures. More specifically, we want to study how the choice of $n$ changes when $A$ changes starting from the FOC

$$wu'(wn + A) = g'(n)$$

We follow the recipe:

1. We write down the FOC as an implicit function (everything on the same side!) and compute the derivatives of

the endogenous variable ($n$)

$$\begin{aligned} wu'(wn+A) - g'(n) &= 0 \\ \frac{\partial wu'(wn+A) - g'(n)}{\partial n} &= w^2u''(wn+A) - g''(n) \end{aligned}$$

2. the exogenous variable is $A$ and the endogenous one is $n$
3. We check the assumptions
   - We assume the function and derivates to be continuous in the domain (we need to assume it because we don't know their functional form)
   - We will apply the IFT to all the optimal points where the FOC holds ($wu'(wn+A)-g'(n) = 0$) but not to those where the partial derivative wrt the endogenous variable is equal to zero $w^2u''(wn+A) - g''(n) = 0$
4. We apply the IFT:

$$F'(x) = \frac{dn}{dA} = -\frac{\frac{\partial wu'(wn+A)-g'(n)}{\partial A}(x,y)}{\frac{\partial wu'(wn+A)-g'(n)}{\partial n}(x,y)} = -\frac{wu''(wn+A)}{w^2u''(wn+A)-g''(n)}$$

Which is the same result we got in the Total Derivative Example. □

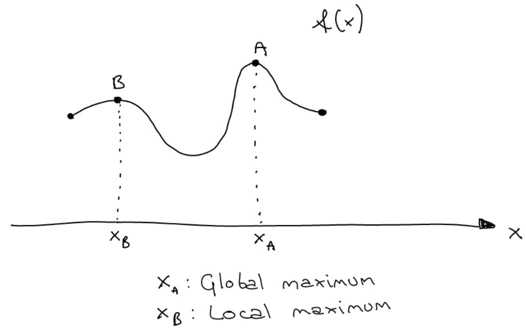
#### 2.3.3 More remarks

Note that the role of the variables $(x, y)$ can be inverted, it is simply necessary to satisfy all the assumptions. This implied that the IFT can be applied to all the points $(x_0, y_0)$ such that $G(x_0, y_0) = 0$ and where at least one of the two partial derivatives is different from zero ($\frac{\partial G(x_0,y_0)}{\partial y} \neq 0$ or $\frac{\partial G(x_0,y_0)}{\partial x} \neq 0$). In case only one of the derivatives is different from zero, the variable with the derivative different from zero is the one you can explicit:

- if $\frac{\partial G(x_0,y_0)}{\partial y} \neq 0$, then we solve for $y$, $y = F(x)$
- if $\frac{\partial G(x_0,y_0)}{\partial x} \neq 0$, then we solve for $x$, $x = \hat{F}(y)$

# 3 Lecture 2

In economics, we are often interest in finding the maxima or minima of functions. For example, households maximize their utility and firms their profits. Recall that a function $f$ has a *local maximum* at a point $x^*$ if $f(x) \leq f(x^*)$ for all $x$ in some open interval containing $x^*$. The maximum is instead *global or absolute* if the unequality holds for all $x$ in the domain of $f$. Both local and global maximum can be *strict* if the inquality is strict. A minimum can be similarly defined.




What about functions of several variables? More formally, what happens if a function is defined from a domain U which is a subset of $\mathbb{R}^n$ to $\mathbb{R}^1$? ($F : U \in \mathbb{R}^n \to \mathbb{R}^1$)? The maximum is similarly defined,$F$ has a local max at a point $\mathbf{x}^*$ if $F(\mathbf{x}^*) \geq F(\mathbf{x})$ for all $\mathbf{x}$ in some open ball containing $\mathbf{x}^*$. The maximum is instead global if the unequality holds for all the $\mathbf{x}$ in the domain (in $U$). For now, we will not introduce balls, but be aware that they are the multivariate counterpart of an interval.

A general optimization problem can be written as

$$\begin{aligned} \text{maximize} \quad & F(x, y) \\ \text{subject to} \quad & h(x, y) = 0 \\ & g(x, y) \geq 0 \\ & x \in \Omega \subset \mathbb{R}, \\ & y \in \hat{\Omega} \subset \mathbb{R}, \end{aligned}$$

where we aim to maximize the function $F$. Here, $\Omega$ and $\hat{\Omega}$ are the domains of the variable of interest, $x$ and $y$ respectively. The condition $h(x, y) = 0$ represents equality constraints, and $g(x, y) \geq 0$ inequality constraints. For simplicity, we consider only one constraint of each type.

How do we solve the above optimization problem? The method depends on the presence and type of constraints:

- if we have a maximization problem without any constraint, we can use the derivative of the function to find the optimal points.
- If there is an equality constraint, we can either use substitution or the Lagrange multiplier method (Lagrangian approach).
- If there is an inequality constraint, we need to use the Kuhn-Tucker conditions (also known as the Karush-Kuhn-Tucker, or KKT, conditions).

Why have we only discussed the maximization of a function and not the minimization?

This is because maximization and minimization are related by the following condition:

$$min(f(x)) \text{ s.t. } g(x) \leq 0 = max(-f(x)) \text{ s.t. } -g(x) \geq 0$$

Thus, it is always possible to transform a minimization problem into a maximization one by negating the function.

### 3.0.1 Unconstrained optimization in $\mathbb{R}$

The simplest case is the absence of any constraint. In such situation, the max of a function can occur

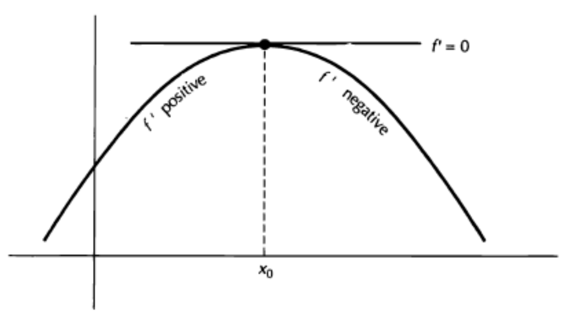
- at the points where the derivative is zero.
- at the points where the derivative cannot be computed [3].
- an endpoints of the domain of $f$.

How do we find such a point? We have a three steps approach

1. **First Order Condition (FOC)**: take the derivative of a function and find the point/s where it is equal to zero (known as *critical* point/s and denoted as $x^*$)
2. **Second Order Condition (SOC)**: verify that the function is *concave* at the critical point by checking if $f''(x^*) \leq 0$.
3. Evaluate the function at the critical point/s and **compare** it/them with the values of the function at the points where the derivative cannot be computed and at the endpoints of the domain

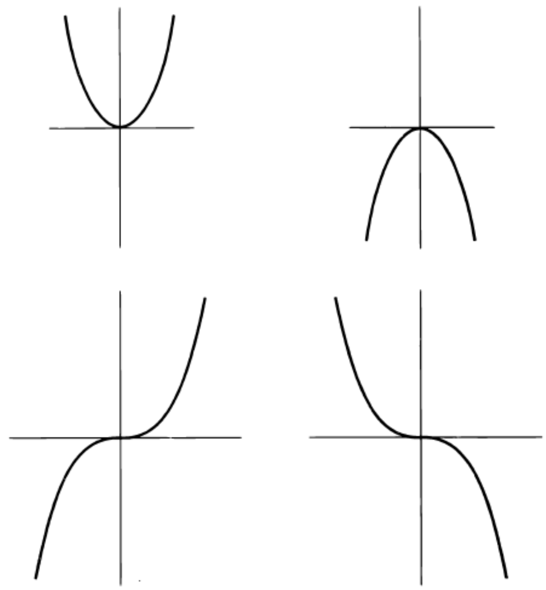
Note that in Economics we often work with functions that are continuous and differentiable, therefore mostly skipping the third step.

Why are we interested in the point where the derivative is null? Because at the max, a function is neither increasing nor decreasing, so its derivative must be null (or undefined or the function must be at an endpoint of the domain). Graphically:




Why check for concavity? To ensure that the critical point is a maximum rather than a minimum or a saddle point. The following plot shows various critical points, but only the concave curve in the top-right has a maximum at the critical point:

[3] In Economics usually we work with differentiable functions, but you might encounter the absolute value function or piecewise functions.



Given that the maximum is a critical point, how do we know if the max is a **global max**?

- if $f$ has only one critical point.
- if $f'' < 0$ throughout *the whole domain*
- when the domain is compact (Weierstrass theorem)

*Example*: maximization of

$$\max_{c_1}(-(c-5)^2+3)$$

We take the FOC and the SOC:

$$\begin{aligned}&\text{FOC: } -2(c-5)=0 \Rightarrow c=5\\&\text{SOC: } -2<0\end{aligned}$$

There is a maximum at $c = 5$. Since the function is defined in all $\mathbb{R}$ and the SOC holds strictly in all the domain, the max is global. □

### 3.0.2 Unconstrained optimization in $\mathbb{R}^n$

Unconstrained optimization for functions of two variables is very similar.

1. **First Order Condition**: take all the partial derivatives of the function and find the interior point/points in which they are all equal to zero (so-called *critical* point/s)

$$\frac{\partial F}{\partial x_i}(\mathbf{x}^*) = 0 \text{ for } i = 1, ..., n$$

2. **Second Order Condition**: we need to study the second derivative to ensure that the critical point is a maximum.

- If the Hessian of the Function evaluated at the critical point is *negative definite* (more on this in the last lecture), then the critical point is a strict local maximum (sufficient condition[4])
- If the Hessian of the function evaluated at the critical point is *negative semidefinite* (more on this in the last lecture) and not negative definite, then the critical point ***may*** be a local maximum. (necessary condition[5])
- If the Hessian of the function evaluated *in a whole ball* around a critical point is *negative semidefinite*, then the critical point is a local maximum.

3. Evaluate the function at the critical point and **compare** it with the value of the function at non-differentiable points and at boundary points.

How do we know if the max is a **global max**? Two sufficient conditions to be simultaneously satisfied for a point to be a maximum are

- if the domain is convex and open. The definition of convex and open will not be discussed in this class, for now be aware that $R^n$ and $\{(x, y)|x > 0 \text{ and } y > 0\}$ are examples of open and convex sets,
- and if the Hessian is negative semidefinite across the entire domain.

### 3.0.3 Constrained optimization: Lagrange approach

Consider the case with two variables and one equality constraint:

$$\begin{aligned} \text{maximize} \quad & F(x, y) \\ \text{subject to} \quad & h(x, y) = 0 \\ & x \in \Omega \subset \mathbb{R}, \\ & y \in \hat{\Omega} \subset \mathbb{R}, \end{aligned}$$

The idea behind the approach is to stransform the constraint problem in an unconstraint one by including a new, "artificial" variable $\lambda$, called the Lagrange multiplier.

To find the maximum of an optimization problem subject to an equality constraint we exploit the Lagrange theorem.

For the Lagrange Theorem to hold the following assumptions needs to hold

- The domain of the functions is a set $U \equiv \Omega \times \Omega \subset \mathbb{R}^n$ that is *open* and *convex*
- A function $f(x, y) : U \to \mathbb{R}$, that is continuous, differentiable and (strictly) concave
- A constraint $h(x, y) : U \to \mathbb{R}$, that is continuous, differentiable and concave
- The critical point $(x^*, y^*)$ (see later) is not a critical point of the constraint $(\nabla h(x^*, y^*) \neq 0)$ [a]

[a] This assumption is called constraint qualification and in case of multiple constraints it would become an assumption of having the gradient of the constraints linearly independent.

[4] This is sufficient because, if satisfied, guarantee that a point is a maximum, but this proof does not go both ways. For example, it is not true that all local maxima have negative definite Hessian matrices.

[5] This is necessary because it must be satisfied for a point to be a maximum. However, satisfying these conditions alone does not guarantee that the point is an extremum; it only indicates potential candidates, the point could also be a saddle or a flat point.

According to the **Lagrange Theorem**, then a point $(x^*, y^*)$ is the maximum of the problem above *if and only if* it exist a real number $\lambda^*$ such that $(x^*, y^*, \lambda^*)$ is the critical point of the Lagrangian function

$$L(x, y, \lambda) \equiv f(x, y) - \lambda(h(x, y))$$

Note that if the function $f(x, y)$ is strictly concave then the maximum is unique.

This is the Theorem in mathematical language, what does it mean more practically? Let's try to write down a recipe to apply. We start by checking the assumptions.

**Assumptions' check**

1. Check if the domain is open and convex. $\mathbb{R}^n$ is an open and convex set; often, we will work with it, or we will rewrite the problem in order to work with it. For example is $\Omega = \mathbb{R}$, then we get $U = \mathbb{R} \times \mathbb{R} = \mathbb{R}^2$
2. Check if the function $f(x, y)$ is continuous and differentiable. To do so, check if the function and its gradient are continuous. Usually, you will work with well-behaved functions.
3. Check if the function $f(x, y)$ is concave or strictly concave. To do so, check if the Hessian of the matrix is negative definite or negative semidefinite (more on this in the last Lecture). (Note that this step is equivalent to the SOC in the unconstrained maximization step)
4. If the constraint is linear, skip this step. Otherwise, check if the function $h(x, y)$ is continuous and differentiable. To do so, check if the function and its gradient are continuous.
5. If the constraint is linear, skip this step. Otherwise, check if the function $h(x, y)$ is concave. To do so, check if the Hessian is negative semidefinite (more on this in the last Lecture).
6. If the constraint is linear, skip this step. Otherwise, compute the critical point for the constraint

$$\nabla h(x_h^*, y_h^*) = 0$$

   - if the point $(x_h^*, y_h^*)$ does not satifies the constraint, discard it
   - if the point $(x_h^*, y_h^*)$ satisfies the constraint, keep it
   - if the point does not exists, nothing happens

**Theorem Application**

1. Construct the Lagrangian function $(L(x, y, \lambda))$, that is compute

$$L(x, y, \lambda) = f(x, y) - \lambda(h(x, y))$$

2. **FOC:** Find the critical points of the lagrangian function by taking all the partial derivatives of the Lagrangian Function and finding the point/s in which they are all equal to zero

$$\frac{\partial L}{\partial x}(x^*, y^*, \lambda^*) = 0 \text{ and } \frac{\partial L}{\partial y}(x^*, y^*, \lambda^*) = 0 \text{ and } \frac{\partial L}{\partial \lambda}(x^*, y^*, \lambda^*) = 0$$

3. Compare the selected maximum with the critical point of the constraint function, if applicable. To do so, compute the value of the function in both points and select the biggest.

(*skip this paragraph if you have never heard about the Bordered Hessian Matrix*) You might be used from past math courses to work with what is called Bordered Hessian matrix (if not, don't worry and skip to the next paragraph!), this is something similar to the SOC and it combines the check on the concavity of $f(x, y)$ with the check on the concavity of $g(x, y)$. More specifically, given m (for us $m = 1$) as the number of constraints and n (for us $n = 2$) the number of variables, if the last $n - m$ leading pricipal minors (more on this in the last lecture) of the Bordered matrix at the critical point alternate the sign starting from $(-1)^{(m+1)}$, the solution is a maximum. In our case if the $2 - 1 = 1$ leading pricipal minor is positive ($sign((-1)^{(1+1)})$).

*Example*: Let's solve

$$\begin{aligned} \text{maximize} \quad & -x - 2y^2 \\ \text{subject to} \quad & x + y = 1 \end{aligned}$$

**Assumptions' check**

1. We are not specifing the domain so we assume that is $\mathbb{R}^2$ that is open and convex.
2. $-x - 2y^2$ is continuous since its the sum of two continuous functions. To check if differentiable, we check if its partial derivatives exist and are continuous. We compute the gradient:

$$\frac{\partial f}{\partial x} = -1$$
$$\frac{\partial f}{\partial x} = -4y$$

   which are both continuous.
3. Check if the function $-x - 2y^2$ is concave or strictly concave. For now assume so.
4. The constraint is linear, we skip this step.
5. The constraint is linear, we skip this step.
6. The constraint is linear, we skip this step.

**Theorem Application**

1. Construct the Lagrangian

$$L(x, y, \lambda) = -x - 2y^2 - \lambda(x + y - 1)$$

2. FOC:

$$\frac{\partial L}{\partial x}(x^*, y^*, \lambda^*) = -1 - \lambda = 0$$
$$\frac{\partial L}{\partial y}(x^*, y^*, \lambda^*) = -4y - \lambda = 0$$
$$\frac{\partial L}{\partial \lambda}(x^*, y^*, \lambda^*) = -(x + y - 1) = 0$$

   From the first equation we get $\lambda = -1$, from the second $y = \frac{3}{4}$, from the last one $x = \frac{1}{4}$. Therefore the critical point is at $x = \frac{1}{4}, y = \frac{3}{4}, \lambda = -1$.

3. Not applicable

The max is the point $(\frac{1}{4}, \frac{3}{4})$.

#### 3.0.4 Constrained optimization: K-T

Consider the case with two variables and one inequality constraint:

$$\begin{aligned} \text{maximize} \quad & F(x, y) \\ \text{subject to} \quad & g(x, y) \geq 0 \\ & x \in \Omega \subset \mathbb{R}, \\ & y \in \hat{\Omega} \subset \mathbb{R}, \end{aligned}$$

To find the maximum of an optimization problem subject to an equality constraint we exploit the Kuhn-Tacker Theorem.

For the K-T Theorem to hold the following assumptions needs to hold

- The domain of the functions is a set $U \equiv \Omega \times \Omega \subset \mathbb{R}^n$ that is *open* and *convex*
- A function $f(x, y) : U \to \mathbb{R}$, that is continuous, differentiable and (strictly) concave
- A constraint $g(x, y) : U \to \mathbb{R}$, that is continuous, differentiable and concave
- A point of the domain exists where the constraint is not binding. In other words, it exists $(x_0, y_0) \in U$ such that $g(x_0, y_0) > 0$

According to the **K-T Theorem**, then a point $(x^*, y^*)$ is the maximum of the problem above *if and only if* it exist a real number $\lambda^*$ such that $(x^*, y^*, \lambda^*)$ satisfies the following conditions (where $L(x, y, \mu) = f(x, y) + \lambda(g(x, y))$ is the Lagrangian):

$$\begin{aligned}
\frac{\partial L}{\partial x}(x^*, y^*, \lambda^*) &= 0 \\
\frac{\partial L}{\partial y}(x^*, y^*, \lambda^*) &= 0 \\
\lambda^*(g(x^*, y^*)) &= 0 \\
\lambda^* &\geq 0 \\
g(x^*, y^*) &\geq 0
\end{aligned}$$

Note that if the function $f(x, y)$ is strictly concave then the maximum is unique.

This is the Theorem in mathematical language, what does it mean more practically? Let's try to write down a recipe to apply. We start by checking the assumptions.

**Assumptions' check**

1. Check if the domain is open and convex. $\mathbb{R}^n$ is an open and convex set; often, we will work with it, or we will rewrite the problem in order to work with it. For example is $\Omega = \mathbb{R}$, then we get $U = \mathbb{R} \times \mathbb{R} = \mathbb{R}^2$
2. Check if the function $f(x, y)$ is continuous and differentiable. To do so, check if the function and its gradient are continuous. Usually, you will work with well-behaved functions.
3. Check if the function $f(x, y)$ is concave or strictly concave. To do so, check if the Hessian of the matrix is negative definite or negative semidefinite (more on this in the last Lecture). (Note that this step is equivalent to the SOC in the unconstrained maximization step)
4. If the constraint is linear, skip this step. Otherwise, check if the function $g(x, y)$ is continuous and differentiable. To do so, check if the function and its gradient are continuous.
5. If the constraint is linear, skip this step. Otherwise, check if the function $g(x, y)$ is concave. To do so, check if the Hessian is negative semidefinite (more on this in the last Lecture).
6. Look at the constraint and look for a point where the constraint is not binding.

**Theorem Application**

1. Construct the Lagrangian function $(L(x, y, \lambda))$, that is compute

$$L(x, y, \lambda) = f(x, y) + \lambda(g(x, y))$$

2. **K-T conditions:** Find the points that satisfy the following conditions:

$$\begin{aligned}
\frac{\partial L}{\partial x}(x^*, y^*, \lambda^*) &= 0 \\
\frac{\partial L}{\partial y}(x^*, y^*, \lambda^*) &= 0 \\
\lambda^*(g(x^*, y^*)) &= 0 \\
\lambda^* &\geq 0 \\
g(x^*, y^*) &\geq 0
\end{aligned}$$

3. First assume that $\lambda = 0$, then assume that $\lambda \neq 0$ and solve.

4. Compare and determine which point provide the highest value in the Lagrangian function

(*skip this paragraph if you have never heard about the Bordered Hessian Matrix*) You might be used from past math courses to work with what is called Bordered Hessian matrix (if not, don't worry and skip to the next paragraph!), this is something similar to the SOC, and it combines the check on the concavity of $f(x, y)$ with the check on the concavity of $g(x, y)$. More specifically, given m (for us $m = 1$) as the number of constraints and n (for us $n = 2$) the number of variables, if the last $n - m$ leading pricipal minors (more on this in the last lecture) of the Bordered matrix at the critical point alternate the sign starting from $(-1)^{(m+1)}$, the solution is a maximum. In our case if the $2 - 1 = 1$ leading principal minor is positive $(sign((-1)^{(1+1)}))$.

Compared to the Lagrangian approach, in the second step we replace the condition derived from the derivative of the Lagrangian with respect to the Lagrangian multiplier, with the slackness condition ( $\lambda^*(g(x^*, y^*)) = 0$) a condition that implies that either one or both parts are equal to zero. Here, if the constraint is binding, that is it holds with equality $(g(x^*, y^*) = 0)$, the maximum is on the constraint and it does not tell us something about $\lambda$. If, instead, the constraint does not bind, it must be that the multiplier is zero.

*Example*: Let's solve

$$\begin{aligned}
\text{maximize} \quad & -x^2 - y^2 + 4x + 6y \\
\text{subject to} \quad & 0 \geq +2x + y - 3
\end{aligned}$$

The first thing to do is to rewrite the constraint as

$$-2x - y + 3 \geq 0$$

Now we follow the step above.

**Assumptions' check**

1. The domain is $\mathbb{R}^2$ which is open and convex

2. $-x^2 - y^2 + 4x + 6y$ is continuous since it is the sum of two continuous functions. To check if differentiable, we

check if its partial derivatives exist and are continuous. We compute the gradient:

$$\frac{\partial f}{\partial x} = -2x + 4$$
$$\frac{\partial f}{\partial x} = -2y + 6$$

which are both continuous.

3. Check if the function $-x^2 - y^2 + 4x + 6y$ is concave or strictly concave. The function is concave in the domain.

4. $-2x - y + 3$ is linear so ok.

5. $-2x - y + 3$ is linear so ok.

6. The point $(0, 0)$ belongs to the domain and satisfies the constraint. So ok.

**Theorem Application**

1. Construct the lagrangian

$$L(x, y, \lambda, \mu) = -x^2 - y^2 + 4x + 6y + \lambda(-2x - y + 3)$$

2. K-T conditions:

$$\frac{\partial L}{\partial x}(x^*, y^*, \lambda^*, \mu^*) = -2x + 4 - 2\lambda = 0$$
$$\frac{\partial L}{\partial y}(x^*, y^*, \lambda^*, \mu^*) = -2y + 6 - \lambda = 0$$
$$\lambda^*(g(x^*, y^*)) = \lambda(-2x - y + 3) = 0$$
$$\lambda^* \geq 0$$
$$g(x^*, y^*) = -2x - y + 3 \geq 0$$

3. 
   - Assume $\lambda = 0$, we get $2x + 4 = 0$ and $-2y + 6 = 0$, so $x = 2$ and $y = 3$, but the K-T condition on the constraint is not satisfied
   - Assume $\lambda\, 0$. We get $-2x - y + 3 = 0$, $-2x + 4 - 2\lambda = 0$ and $-2y + 6 - \lambda = 0$, so that from the first we have $y = -2x + 3$, which we plug in the third one and get $4x - 6 + 6 - \lambda = 0$ and $\lambda = 4x$, we plug it in the second one and get $-2x + 4 - 8x = 0$, and finally we have $x = 2/5, y = 11/5, \lambda = 8/5$, which satisfies all the conditions.

   The point

$$(\frac{2}{5}, \frac{11}{5})$$

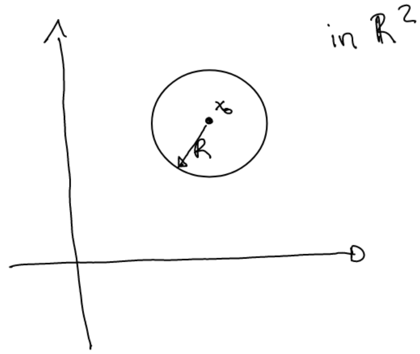
   is the solution of the maximization problem.

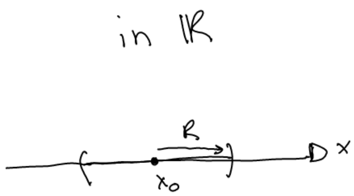
#### 3.0.5 APPENDIX TO THE FIRST DAY

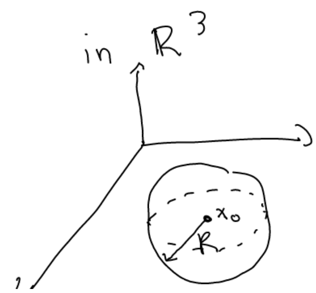
In this Appendix I will again introduce some of the topics discussed in class but there were not on the slides.

**Ball**

A ball is simply a ball, let's look at the pot at the top below. In $\mathbb{R}^2$, a ball of center $x_0$ and radius $R$, is the set of points inside the circle that has center in $x_0$ and radius $R$ (note that it is composed only by the points inside! it does not include its border, it does not include the circumference). In $\mathbb{R}$, a ball is an open interval, while in $\mathbb{R}^3$ is a sphere.




What do we write mathematically?

$$B(x_0, R) = \{x \in \mathbb{R}^n : |x - x_0| < R\}$$

and we read it as a ball centered in $x_0$ of radius R is the set of points belonging to $\mathbb{R}^n$ such that the distance between the point and $x_0$ is smaller (strictly!) than the radius.

**Open Set**

There are several definitions of open sets, the one we consider here tells us that a set is open if in every point of the domain, we can draw a ball that is included in the domain. Example of open sets are $\mathbb{R}^n$ and $\{(x, y) : x > 0, y > 0\}$.

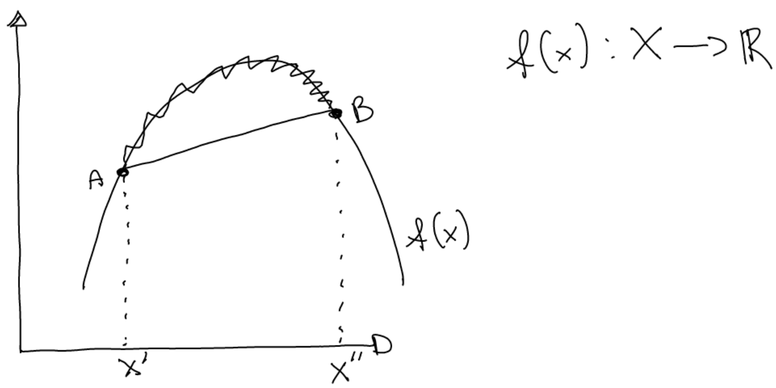
**Convex Set**

There are several definitions of convex sets, the one we consider here tells us that a set is convex if given any two points in the set, the segment that connects the two points belongs to the set too. Examples are $\mathbb{R}^n$ and the set below on the right:


In fact on the right the segment AB belongs to the set too, on the left we have a nonconvex set since the segment CD does not belong to the set.

**Concave Function**

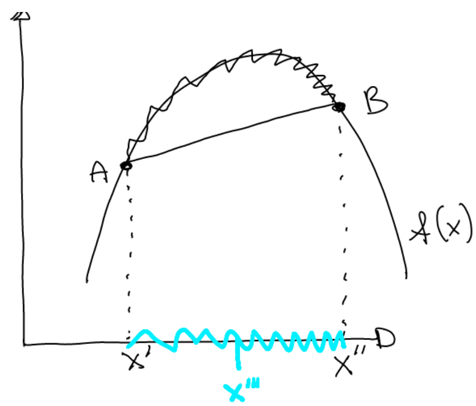
We all know how does a convex function looks like, but let's try to understand its definition. Let's plot a convex function:




A function is concave if, given any two points on the function, the segment connecting the two points is below the arch connecting them. Here clearly the segment that connect A and B is below the arch that connects them. Mathematically, we write given $x', x'' \in X$, where $X$ is the domain of $f$, and given a variable $\lambda \in [0, 1]$, a function is concave if the following condition holds for any two points and any lambda:

$$f(\lambda x' + (1 - \lambda)x'') \geq \lambda f(x') + (1 - \lambda)f(x'')$$

Why? Let's try to study what we wrote. We start by $\lambda x' + (1 - \lambda)x''$, if $\lambda = 0$ then this value becomes $x'$, if $\lambda = 1$ then this value becomes $x''$, if $\lambda$ is between 0 and 1, then we get any of the point on the segment that connects $x'$ and $x''$ on the x-axis (in light blu in the plot):




Ok so $\lambda x' + (1-\lambda)x''$ is the segment in light blu, what is $f(\lambda x' + (1-\lambda)x'')$? Is the function evaluated at the segment, that is the y values of the arch connecting A and B. Similarly, $\lambda f(x') + (1-\lambda)f(x'')$ are the y values of the segment connecting A and B.

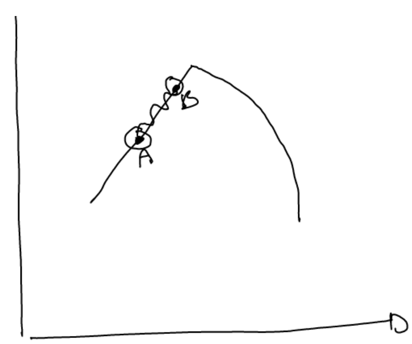
**Strictly Concave Function**

If the above condition holds strictly, we have a strictly concave function. That is given $x', x'' \in X$, where $X$ is the domain of $f$, and given a variable $\lambda \in (0, 1)$, a function is concave if the following condition holds for any two points and any lambda:

$$f(\lambda x' + (1-\lambda)x'') > \lambda f(x') + (1-\lambda)f(x'')$$

**Strictly Concave vs Concave Function**

Let's see as an example of function that is concave but not strictly concave:




In fact, here the arch connecting A and B is not above the segment, it coincides with it! Therefore the above inequality does not hold in a strict sense.

**Necessary and sufficient condition in K-T Theorem**

The version of the theorem seen in class is a ”nice” version; that is, our assumption guarantees that we can use the ”if and only if” in the theorem, they guarantee that it is both a sufficient and a necessary condition.

Let’s have a review of necessary and sufficient condition

if assumption holds

$$\begin{tikzcd} x^* \text{ is a max} \arrow[r, bend left, "\text{necessary}"] & \text{KKT conditions are satisfied} \arrow[l, bend left, "\text{sufficient}"] \end{tikzcd}$$

so the necessary condition is such that if a point is a maximum we have that the KKT conditions are satisfied, but the opposite might not be true. A sufficient condition is a condition that tells you that if the KKT conditions are satisfied, then we have a maximum but the opposite might not be true.

The assumption seen in class guarantee that the KKT condition is both necessary and sufficient for the maximum.

What happens if we relax some assumtions? Remember that our assumptions were:

- The domain of the functions is a set $U \equiv \Omega \times \Omega \subset \mathbb{R}^n$ that is *open* and ***convex***
- A function $f(x, y) : U \to \mathbb{R}$, that is continuous, differentiable and **(strictly) concave**
- A constraint $h(x, y) : U \to \mathbb{R}$, that is continuous, differentiable and **concave**
- **A point of the domain exists where the constraint is not binding. In other words, it exists** $(x_0, y_0) \in U$ **such that** $g(x_0, y_0) > 0$

we can relax the assumption in bold, replace the last assumption and get

- The domain of the functions is a set $U \equiv \Omega \times \Omega \subset \mathbb{R}^n$ that is *open*
- A function $f(x, y) : U \to \mathbb{R}$, that is continuous, differentiable
- A constraint $h(x, y) : U \to \mathbb{R}$, that is continuous, differentiable
- The gradients of the active constraints are linearly independent.

These new assumptions are less restrictive then before. Under the new assumptions, the KKT conditions are NOT any more sufficient; they are only necessary. So we know they are verified in a maximum, but we cannot use them to find a maximum.

# 4 Lecture 3

In this lecture, we will explore a dynamic continuous choice problem. The goal is to introduce the setup of the dynamic programming approach, with solution methods to be covered in future courses.

Consider a scenario where you have a cake of size $W_0$. At each point of time, $t = 0, 1, 2, 3, \ldots .T$ you can consume some of the cake and thus save the remainder. Let $c_t$ be your consumption in period t and let $u(c_t)$ represent the flow of utility from this consumption. Assume $u(\cdot)$ is real-valued, differentiable, strictly increasing and strictly concave[6]. Assume that $\lim_{c \to 0} u'(c) \to \infty$. Represent lifetime utility by

$$\sum_{t=0}^{T} \beta^{(t)} u(c_t)$$

where $0 < \beta < 1$ and $\beta$ is called the discount factor.

The cake does not depreciate (melt) or grow. Hence, the evolution of the cake over time is governed by:

$$W_{t+1} = W_t - c_t$$

for $t = 0, 1, 2, ..T$. How would you find the optimal path of consumption, $\{c_t\}_0^T$ ?

The standard approach would be to solve the following constrained optimization problem (called the **sequence problem** because we are looking for sequences of numbers)

$$\begin{aligned} \max_{\{c_t\}_0^T, \{W_t\}_1^{T+1}} & \sum_{t=0}^{T} \beta^{(t)} u(c_t) \\ \text{subject to } W_{t+1} &= W_t - c_t \text{ for } t = 0, 1, 2, 3, ...T \\ c_t &\geq 0 \text{ for } t = 0, 1, 2, 3, ...T \\ W_t &\geq \text{ for } t = 0, 1, 2, 3, ...T \end{aligned}$$

The dynamic programming approach reformulates the above problem and converts a $T$ period problem into a 2 period problem with the appropriate rewriting of the objective function, making it easier to work on it.

## 4.1 Finite Horizon Problem

We will see now how the cake eating problem can be reformulated in dynamic programming.

Consider the following. Finding the optimal path of consumption implies acting optimally and finding the optimal consumption at period 0, at period 1, etc. If you are acting optimally, you only need to focus on choosing the optimal level of consumption at the current period, since future decisions will also be made optimally. Of course, today's consumption decision affects future consumption possibilities, but you do not need to explicitly plan the entire future consumption path today. You can rely on the fact that you will continue to behave optimally and maximize your utility in the future. With this in mind, we can reformulate the problem as follows.

[6]Additionally, the utility function is not indexed by time: preferences are stationary.

Given $W_0$, the problem becomes

$$\max_{c_0} u\left(c_0\right) + \beta V_T\left(W_1\right)$$
$$\text{subject to } W_1 = W_0 - c_0$$

In this formulation, the choice of consumption in period 0 determines the size of the cake that will be available starting in period $1, W_1$. So instead of choosing a sequence of consumption levels, we focus on just choosing $c_0$. Once $c_0$ and thus $W_1$ are determined, the value of the problem from then on is given by $V_T\left(W_1\right)$, where $T$ is the number of horizon left and $W_1$ is the size of the cake left at $t=1$. $V_T\left(W_1\right)$ represents the maximal utility flow from a $T$ period problem given a size $W_1$ cake and is known as **value function**. This function completely summarizes optimal behavior from period 1 onwards. For the purposes of the dynamic programming problem, it doesn't matter how the cake will be consumed after the initial period. All that is important is that the agent will be acting optimally and thus generating utility given by $V_T\left(W_1\right)$. This is the *principle of optimality*, due to Richard Bellman, at work. With this knowledge, an optimal decision can be made regarding consumption in period 0. Note that the value of utility of $V_T\left(W_1\right)$ is the value at time $t=1$, to find its value at $t=0$ we multiply it by the discount factor $\beta$.

*Example*: To better understand this formulation, consider a problem with three periods: $t = 0, 1, 2$. At time 0, you receive utility from current consumption $u(c_0)$ and choose $c_0$ with the understanding that future decisions about $c_1$ and $c_2$ will also be made optimally. The value of future utility, given optimal choices, is $V_2 = [u(c_1^*) + \beta u(c_2^*)]$. The optimal future choices depend on the remaining cake size $W_1$, so we denote it as $V_2(W_1) = [u(c_1^*) + \beta u(c_2^*)]$. At time 1, the problem is similar but only one period remains, so the decision involves $c_1$ and $V_1(W_2) = [u(c_2^*)]$. At time 2, you simply choose $c_2$. □

The value of the abovementioned problem is denoted as $V_{T+1}(W_0)$:

$$V_{T+1}(W_0) \equiv \max_{c_0} u\left(c_0\right) + \beta V_T\left(W_1\right)$$
$$\text{subject to } W_1 = W_0 - c_0 \text{ and given } W_0$$

Now it should be more clear that

$$V_T(W_1) \equiv \max_{c_1} u\left(c_1\right) + \beta V_{T-1}\left(W_2\right)$$
$$\text{subject to } W_2 = W_1 - c_1 \text{ and given } W_1$$

$$V_{T-1}(W_2) \equiv \max_{c_2} u\left(c_2\right) + \beta V_{T-2}\left(W_3\right)$$
$$\text{subject to } W_3 = W_2 - c_2 \text{ and given } W_2$$

and so on.

To solve the above finite horizon problem we have to find the value of $V_T\left(W_1\right)$, a possible solution method is to solve the problem backward, but discussing this method in detail is beyond the scope of this lecture.

## 4.2 Infinite Horizon Problem

Suppose that we consider the above problem and allow the horizon to go to infinity. As before, one can consider solving the infinite horizon sequence problem given by:

$$\max_{\{c_t\}_0^\infty, \{W_t\}_1^\infty} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

along with the transition equation of

$$W_{t+1} = W_t - c_t$$

for t $= 0, 1, 2, \ldots\ldots$

Specifying this as a dynamic programming problem (or **recursive problem**),

$$V(W) = \max_{c \in [0,W]} u(c) + \beta V(W - c)$$

for all $W$.

Here $u(c)$ is again the utility from consuming c units in the current period. $V(W)$ is the value of the infinite horizon problem starting with a cake of size $W$. So in the given period, the agent chooses current consumption and thus reduces the size of the cake to $W' = W - c$, as in the transition equation. We use variables with primes to denote future values. The value of starting the next period with a cake of that size is then given by $V(W - c)$ which is discounted at rate $\beta < 1$.

Now we will introduce some terminology. For this problem,

- the **state variable** is the size of the cake $(W)$ that is given at the start of any period. The state completely *summarizes all information from the past* that is needed for the forward looking optimization problem.
- the **control variable** is the variable that is being chosen. In this case, it is the level of consumption in the current period, $c$. Note that $c$ lies in a compact set.
- the **transition equation** describes the dependence of the state tomorrow on the state today and the control today. It corresponds to $W' = W - c$

Alternatively, we can specify the problem so that instead of choosing today's consumption we choose tomorrow's state.

$$V(W) = \max_{W' \in [0,W]} u(W - W') + \beta V(W')$$

for all $W$. Either specification yields the same result. But choosing tomorrow's state often makes the algebra a bit easier so we will work with the latter.

This expression is a functional equation, since the unknown is a function $V(W)$, and is often called a **Bellman equation** after Richard Bellman, one of the originators of dynamic programming. Note that the *unknown* in the

Bellman equation is the *value function* itself: the idea is to find a function $V(W)$ that satisfies this condition for all $W$. Unlike the finite horizon problem, there is no terminal period to use to derive the value function. In effect, the fixed point restriction of having $V(W)$ on both sides of the equation will provide us with a means of solving the functional equation.

Note too that time itself does not enter into Bellman's equation: we can express all relations without an indication of time. This is the essence of *stationarity*. In fact, we will ultimately use the stationarity of the problem to make arguments about the existence of a value function satisfying the functional equation.

A final very important property of this problem is that all information about the past that bears on current and future decisions is summarized by $W$, the size of the cake at the start of the period. Whether the cake is of this size because we initially had a large cake and ate a lot or a small cake and were frugal is not relevant. All that matters is that we have a cake of a given size. This property partly reflects the fact that the preferences of the agent do not depend on past consumption. But, in fact, if this was the case, we could amend the problem to allow this possibility.

In the future classes, you will address the question of whether there exists a value function that satisfies the Bellman equation. For now, we assume that a solution exists and explore its properties.

### 4.2.1 First Order Condition

We derive w.r.t. $W'$ and we get the following FOC:

$$u'(c) = \beta V'(W')$$

This looks simple but what is the derivative of the value function?

This seems particularly hard to answer since we do not know $V(W)$. However, to answer this question suppose we know the solution to the problem and denote it via $g_V(W)$:

$$g_V(W) = argmax_{W'} u(c) + \beta V(W')$$

We rewrite the problem above as

$$V(W) = u(W - g(W)) + \beta V(g(W))$$

we take the derivative (assuming the value function is differentiable)

$$\begin{aligned} V'(W) &= [1 - g'(W)]u'(c) + \beta g'(W)V'(W') \\ &= u'(c) + g'(W)[\beta V'(W') - u'(c)] \end{aligned}$$

The terms in the square bracket is zero since the FOC is $u'(c) - \beta V'(W') = 0$ and must hold at all times. Therefore:

$$V'(W) = u'(c)$$

Since this holds for all $W$, it will hold in the following period too:

$$V'(W') = u'(c')$$

By combining the result with the FOC, we get

$$u'(c) = \beta u'(c')$$

This is a necessary condition of optimality for any $t$: if it is violated the agent can do better by adjusting $c_t$ and $c_{t+1}$. Such a condition is reffered to as **Euler Equation**. Essentially, the Euler equation says that the marginal utility cost of reducing consumption by $\varepsilon$ in the current period equals the marginal utility gain from consuming the extra $\varepsilon$ of cake in the next period, which is discounted by $\beta$. If the Euler equation holds, then it is impossible to increase utility by moving consumption across adjacent periods given a candidate solution. It should be clear though that this condition may not be sufficient: it does not cover deviations that last more than one period. For example, could utility be increased by reducing consumption by $\varepsilon$ in the current period saving the "cake" for two periods and then increasing consumption in period in two periods ? Clearly this is not covered by a single Euler equation. However, by combining the Euler equation that hold across periods, the one between the current and the next and the one between the next and in two periods, we can see that such a deviation will not increase utility. This is simply because the combination of Euler equations implies:

$$u'(c) = \beta^2 u'(c_{t+2})$$

so that the two-period deviation from the candidate solution will not increase utility.

### 4.2.2 The Policy Function

The policy function provide a mapping from the state variable to the optimal choice in the control variable. It is a function that describes the optimal behaviour and, in this context, it links the level of consumption, and therefore next period's cake (the controls from the different formulations), to the size of the cake (the state).

$$\begin{aligned} c &= \phi(W) \\ W' &= g(W) = W - \phi(W) \end{aligned}$$

Using these in the Euler equation reduces the problem to these policy functions alone:

$$u'(\phi(W)) = \beta u'(\phi(W - \phi(W)))$$

for all $W$.

## 4.3 Adding a shock

In the dynamic programming approach is also very simple to add uncertainty, for the cake-eating problem we will add a taste shock, that is uncertainty on how much the individual values consumption. The new utility is

$$\epsilon u(c)$$

where $\epsilon$ is a random variable that takes value either low ($\epsilon_L$) or high ($\epsilon_H$), with $\epsilon_H > \epsilon_L > 0$. Assume that the agent knows the value of the taste shock when making current decision but does not know the future values of this shock. Thus the agent must use expectations of future values of $\epsilon$ when deciding how much cake to eat today.

For now assume that the realization of the shock at each period is independent on past realizations. The new cake eating problem is

$$V(W,\epsilon) = \max_{c\in[0,W]} \epsilon u(c) + \beta E V(W',\epsilon')$$

Note that now we added the $\epsilon$ to the state variable since is known to the agent at the moment of the choice and we added the expectation over future realizations, which implies averaging over possible future states, weighted by their probabilities.

Assume that that the realization of the shock at each period does depend on its past values. More specifically, assume $\epsilon$ today depends only on the value of the realization in the previous period, that is assume that $\epsilon$ follows a first-order Markov process. If the value of $\epsilon$ was low in the past period, there is a probability $\pi_{lh} \equiv Prob(\epsilon' = \epsilon_H|\epsilon = \epsilon_L)$ of having a high value of $\epsilon$ and of $\pi_{ll} = Prob(\epsilon' = \epsilon_L|\epsilon = \epsilon_L)$ of having a low one. Similarly, if the value of $\epsilon$ was high in the past period, there is a probability $\pi_{hl} \equiv Prob(\epsilon' = \epsilon_L|\epsilon = \epsilon_H)$ of having a low value of $\epsilon$ and of $\pi_{hh} = Prob(\epsilon' = \epsilon_H|\epsilon = \epsilon_H)$ of having a high one. This information is summarized in the transition matrix:

$$\Pi = \begin{pmatrix} \pi_{ll} & \pi_{lh} \\ \pi_{hl} & \pi_{hh} \end{pmatrix}$$

Now the agent in choosing the consumption needs to take into account that it may be optimal to consume less today (save more) in anticipation of a high realization of $\epsilon$ in the future. The new Bellman equation is

$$V(W,\epsilon) = \max_{c\in[0,W]} \epsilon u(c) + \beta E_{\epsilon'|\epsilon} V(W',\epsilon')$$

Note that the main difference is the presence of the conditional expectation, which summarizes the following two situations:

$$V(W,\epsilon_L) = \max_{c\in[0,W]} \epsilon u(c) + \beta[\pi_{lh} V(W',\epsilon_H) + \pi_{ll} V(W',\epsilon_L)]$$
$$V(W,\epsilon_H) = \max_{c\in[0,W]} \epsilon u(c) + \beta[\pi_{hh} V(W',\epsilon_H) + \pi_{hl} V(W',\epsilon_L)]$$

Following the same procedure as above we can find the FOC:

$$\epsilon u'(W - W') = \beta E_{\epsilon'|\epsilon}[\epsilon' u'(W' - W'')]$$

The policy function will now depend also on the realization of the shock:

$$W' = \varphi(W,\epsilon)$$

## 4.4 Mathematical tools for dynamic programming

*These tools are more advanced, I am not sure if we will cover them in class, but if you have the time feel free to read the next section, otherwise go to Lecture 4*

Let's see some mathematical tools that you will use in more advanced courses of dynamic programming.

### 4.4.1 Metric space

A metric space is a "mathematical object" composed by a set $S$ and a function $d$ called metric. $d$ is such that it respects the rules of the distance, that is for all $x, y, z \in$ the following three conditin are true:

(a) $d(x, y) \geq 0$, with equality iff $x = y$, that is the distance between two points is zero if $x = y$, otherwise must be positive. (it's a simple concept, the distance between two points in the real world is never negative!)

(b) $d(x, y) = d(y, x)$, that is the distance is the same no matter whether you are measuring it from x to y or from y to x

(c) $d(x, y) \leq d(x, z) + d(z, y)$. (To understand this think about the distance between you and one person. You will think about it as the straight line that connect you to this person and, given a third person, this line will be equal or smaller to the same of the distance between x and z and between y and z)

Why do we need metric spaces? Because to solve the Bellman equation above you will make use of converge and you need the notion of distance, and therefore the notion of metric space.

What is d()? An example of d() is the euclidian distance $d((x, y), (x', y')) = \sqrt{(x - x')^2 + (y - y')^2}$.

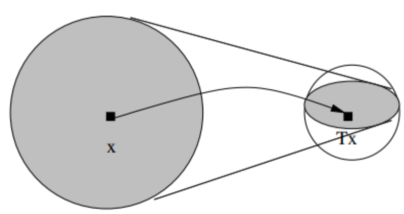
### 4.4.2 Contraction mapping

Let $T : S \to S$ be a operator tht maps the set $S$ to itself. This operator is a contraction mapping if it exist a $\beta \in (0, 1)$ such that

$$d(Tz_1, Tz_2) \leq \beta d(z_1, z_2) \forall z_1, z_2 \in S$$

and $\beta$ is called modulus of the contraction mapping.

What does it mean? The contraction mapping is a function that, if applied to two points, it brings them closer to one another: the distance $d(z_1, z_2)$, once you apply $T$ to each point, becomes smaller (or equal).




An easy example of contraction if $T = \alpha x$ with $\alpha \in (0, 1)$ and $T : \mathbb{R} \to \mathbb{R}$ such for $x \in \mathbb{R}$ you get $\alpha x$.

Sometimes the notation of contraction might be misleading. Suppose for example you are applying a contraction $T$ to a point x, then you might write $Tx$ or $T(x)$. Suppose you are applying $T$ to a function $V(x)$, then I have seen writing $T(V)(x)$ or $(TV)(x)$.

### 4.4.3 Fixed point

When dealing with contractions we are often interested in finding the point $z$ in $M$ such that if you apply to it the contraction $T$, you get the same point again

$$Tz = z$$

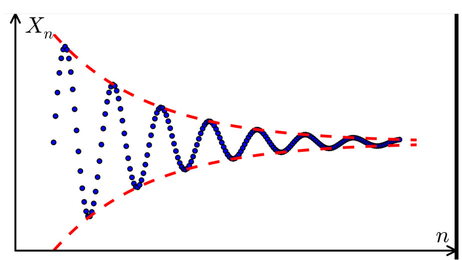
### 4.4.4 Convergent sequence

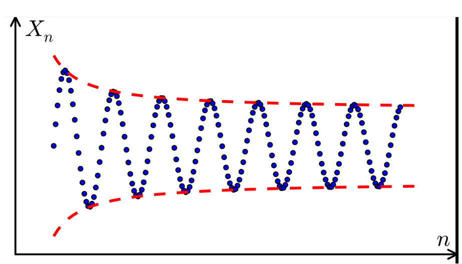
A sequence is a function that goes from $\mathbb{N}$ to a set X and is denoted as $\{x_n\}_{n=0}^{\infty}$. An example is the sequence $n^2$, that associates at each natural number its squared value.

A sequence $\{x_n\}_{n=0}^{\infty}$ converges to $\hat{x} \in S$ if for all $\epsilon > 0$ there exists an $N_\epsilon \in \mathbb{N}$ such that for every $n > N_\epsilon$, $d(x_n, \hat{x}) < \epsilon$. That is you take a positive value $\epsilon$ and the point $\hat{x}$, then it does exist a natural number $N$ and its associated number $x_N$, such that the distance of element in the sequence that come after $x_N$ with the point $\hat{x}$ is smaller than $\epsilon$. Putting it simple, the sequence eventually reaches a certain value as the natural numbers reach infinity.

### 4.4.5 Cauchy sequence

A sequence $\{x_n\}_{n=0}^{\infty}$ is a Cauchy sequence if for all $\epsilon > 0$ there exists an $N_\epsilon \in \mathbb{N}$ such that for every $n, m > N_\epsilon$ with $n, m \in \mathbb{N}$, $d(x_n, x_m) < \epsilon$. A sequence is Cauchy if the points get closer and closer to each other. Roughly speaking, the terms of the sequence are getting closer and closer together in a way that suggests that the sequence ought to have a limit in S. Nonetheless, such a limit does not always exist within S, does not always converge to an element of the set. Example: $(1 + 1/n)^n$ does not converge in $\mathbb{Q}$ (set of rational number) and is Cauchy. Have a look below, the first image displays a cauchy sequence, the second not (because the elements of the sequence do not get arbitrarily close to each other as the sequence progresses.)




It is possible to prove that every convergent sequence is also a cauchy sequence, but the opposite does not hold.

#### 4.4.6 Complete metric space

A metric space is complete if every Cauchy sequence converges to some point in the set of the metric space.

In other words, the idea is that a Cauchy sequence looks like it should be converging to something, and a space is complete if there is always something there for it to converge to.

#### 4.4.7 Final remarks

The mathematical concepts introduced will be necessary for the contraction mapping theorem, the theorem on which is based the value function iteration methods, one of the most famous method used to solve Bellman equations.

## 5 Lecture 4

We will now work with matrices which are simply rectangular arrays of numbers and their properties. We will also see that all the standard operations can be apply to matrices: they can be added, subtracted, multiplied and divided.

What is a matrix?

Given $m, n \in \mathbb{N}$ a matrix of order $m \times n$ is a table of numbers with m rows and n columns as displayed below

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1j} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2j} & \cdots & a_{2n} \\ \vdots & & \ddots & & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & & & \ddots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mj} & \cdots & a_{mn} \end{bmatrix} \tag{2}$$

where each element $a_{ij}$ is a number and is called the $(i, j)$ entry of the matrix.

The standard is to use the first subscript of the entries of a matrix to denote the row the entries belongs to, the second subscript $j$ denotes the column the entries belong to. Usually to refer to a matrix capital letters are used and we will write $A_{m\times n}$ to denote a matrix of order $m \times n$. The main diagonal of a matrix is made up by the entries $a_{ii}$ with $i \in \{1, ..., m\}$

*Example*: An example of a matrix of order $3 \times 4$ is

$$\begin{bmatrix} 1 & 4 & 5 & 8 \\ 34 & 2.3 & 9 & 10 \\ \frac{2}{3} & 0 & 1 & 0 \end{bmatrix}$$

where the main diagonal is made up by $\{1, 2.3, 1\}$ □

We will know study specific types of matrices:

- A **vector** is a matrix of the form
$$\begin{bmatrix} a_1 \\ \cdots \\ a_m \end{bmatrix} \text{ or } \begin{bmatrix} a_1, & \cdots & a_n \end{bmatrix} \tag{3}$$
where the first is called column vector and the second row vector.

- A **square matrix** is a matrix where the number of rows is equal to the number of columns. That is a matrix $A_{m \times n}$ is square if $m = n$.

  *Example*:
$$B_{2\times 2} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \text{ and } C_{3\times 3} = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 4 & 5 & 7 \end{bmatrix}$$
□

- A **diagonal matrix** is a matrix in which the entries outside the main diagonal are all zero

  *Example*:
$$A_{2\times 2} \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix} \text{ and } B_{3\times 3} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 7 \end{bmatrix}$$
□

- The **zero matrix** is a matrix of zeros

  *Example*:
$$\mathbf{0}_{2\times 2} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \text{ and } \mathbf{0}_{3\times 3} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$
□

  Note that the zero matrix is usually denoted by a 0 in bold. If the subscpit is of the type $m \times n$ you have a zero matrix of order $m \times n$, otherwise the subscript could simply be a single natural number ($\mathbf{0}_m$) to indicator a zero vector.

- The **identity matrix** is a square matrix in which all the elements of principal diagonals are one, and all other elements are zeros

  *Example*:
$$\mathbf{I}_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \text{ and } \mathbf{I}_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$
□

  Note that it is denoted by the notation $\mathbf{I}_n$ or simply $\mathbf{I}$.

- A **upper-triangular matrix** is a matrix in which all entries below the diagonal are 0.

  *Example*:
$$B_{2\times 2} = \begin{bmatrix} 1 & 2 \\ 0 & 4 \end{bmatrix} \text{ and } C_{3\times 3} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 7 \end{bmatrix}$$

□

- A **lower-triangular matrix** is a matrix in which all entries above the diagonal are 0.

  *Example*:

$$B_{2\times 2} = \begin{bmatrix} 1 & 0 \\ 3 & 4 \end{bmatrix} \text{ and } C_{3\times 3} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ 4 & 5 & 7 \end{bmatrix}$$

□

## 5.1 Fundamentals of Matrix algebra

### 5.1.1 Addition

It is possible to add two matrices if they are of the same size, that is if they have the same number of rows and columns. Their sum is a new matrix of the same size as the two matrices being added.

How does addition work? The $(i, j)$th entry of the sum matrix is simply the sum of the $(i, j)th$ entried of the two matrices being added:

$$\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & a_{ij} & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} + \begin{pmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & b_{ij} & \vdots \\ b_{m1} & \cdots & b_{mn} \end{pmatrix} = \begin{pmatrix} a_{11}+b_{11} & \cdots & a_{1n}+b_{1n} \\ \vdots & a_{ij}+b_{ij} & \vdots \\ a_{m1}+b_{m1} & \cdots & a_{mn}+b_{mn} \end{pmatrix}$$

*Example*:

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} + \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{pmatrix} = \begin{pmatrix} 1+1 & 2+1 & 3+1 \\ 4+2 & 5+2 & 6+2 \\ 7+3 & 8+3 & 9+3 \end{pmatrix} = \begin{pmatrix} 2 & 3 & 4 \\ 6 & 7 & 8 \\ 10 & 11 & 12 \end{pmatrix}$$

□

### 5.1.2 Subtraction

It is possible to subtract two matrices if they are of the same size, that is if they have the same number of rows and columns. Their subtraction is a new matrix of the same size as the two matrices being added.

How does subtraction work? The $(i, j)$th entry of the sum matrix is simply the subtraction of the $(i, j)th$ entries of the two matrices being added:

$$\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & a_{ij} & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} - \begin{pmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & b_{ij} & \vdots \\ b_{m1} & \cdots & b_{mn} \end{pmatrix} = \begin{pmatrix} a_{11}-b_{11} & \cdots & a_{1n}-b_{1n} \\ \vdots & a_{ij}-b_{ij} & \vdots \\ a_{m1}-b_{m1} & \cdots & a_{mn}-b_{mn} \end{pmatrix}$$

*Example*:

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} - \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{pmatrix} = \begin{pmatrix} 1-1 & 2-1 & 3-1 \\ 4-2 & 5-2 & 6-2 \\ 7-3 & 8-3 & 9-3 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 2 \\ 2 & 3 & 4 \\ 4 & 5 & 6 \end{pmatrix}$$

$\square$

### 5.1.3 Scalar multiplication

It is possible to multiply a scalar by a ordinary number. The product of a matrix $A$ and a scalar $c$, denoted by $cA$, is the matrix created by multiplying each entry of $A$ by $c$.

$$cA = c\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & a_{ij} & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} = \begin{pmatrix} ca_{11} & \cdots & ca_{1n} \\ \vdots & ca_{ij} & \vdots \\ ca_{m1} & \cdots & ca_{mn} \end{pmatrix}$$

*Example*:

$$2A = 2\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} = \begin{pmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \\ 14 & 16 & 18 \end{pmatrix}$$

$\square$

### 5.1.4 Matrix Multiplication

It is also possible to multiply two matrices, but not all matrices can be multiplied and the order of multiplication does matter. It is possible to multiply two matrices $A$ and $B$, only if

$$\text{the number of columns of } A = \text{to the number of rows of } B$$

In other words if, given two matrices $A_{m\times n}$ and $B_{p\times k}$, it is possible to have the product $AB$ only if $n = p$ or to have the product $BA$ only if $k = m$.

What is the matrix product? Each entry of the matrix product $A_{m\times n}B_{n\times k}$ can be obtained by multiplying the $i$th row of $A$ and the $j$th column of $B$ as follows:

$$\begin{pmatrix} a_{i1} & a_{i2} & \cdots & a_{in} \end{pmatrix} \cdot \begin{pmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{nj} \end{pmatrix} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

*Example*:

$$\begin{pmatrix} 1 & 2 \\ 4 & 5 \\ 7 & 8 \end{pmatrix}\begin{pmatrix} 0 & 3 \\ 6 & 9 \end{pmatrix} = \begin{pmatrix} 1\cdot 0 + 2\cdot 6 & 1\cdot 3 + 2\cdot 9 \\ 4\cdot 0 + 5\cdot 6 & 4\cdot 3 + 5\cdot 9 \\ 7\cdot 0 + 8\cdot 6 & 7\cdot 3 + 8\cdot 9 \end{pmatrix} = \begin{pmatrix} 0+12 & 3+18 \\ 0+30 & 12+45 \\ 0+48 & 21+72 \end{pmatrix} = \begin{pmatrix} 12 & 21 \\ 30 & 57 \\ 48 & 93 \end{pmatrix}$$

$\square$

The identity matrix has the following property: if you multiply a matrix with the identity matrix you get the original matrix. That is

$$AI = A \text{ and } IA = A$$

An idempotent matrix is a square matrix that multiplied by itself gives itself ($B \cdot B = B$ such as the identity matrix

*Example*: an idempotent matrix is

$$\begin{pmatrix} 5 & -5 \\ 4 & -4 \end{pmatrix}$$

□

#### 5.1.5 Law of Matrix Algebra

- **Associative Laws**:

$$(A+B)+C = A+B(B+C)$$
$$(AB)C = A(BC)$$

- **Commutative Law for Addition**:

$$A+B = B+A$$

- **Distributive Laws**:

$$A(B+C) = AB+AC$$
$$(A+B)C = AC+BC$$

*Example:* Given A,B,C square matrices with the same number of rows, simplify the following expression using matrix algebra rules:

$$A(BC+\alpha I) - B(2AC - 3\beta I)$$

and arrive at

$$(AB - 2BA)C + \alpha A + 3\beta B$$

Follow the following steps:

1. Distribute $A$ and $B$:

$$ABC + \alpha A - 2BAC + 3\beta B$$

2. Combine like terms involving $AC$:

$$ABC - 2BAC + \alpha A + 3\beta B$$

3. Factor out common terms:

$$(AB - 2BA)C + \alpha A + 3\beta B$$

4. Final simplification:

$$(AB - 2BA)C + \alpha A + 3\beta B$$

□

### 5.1.6 Transpose

The transpose of a $m \times n$ matrix A is the $n \times m$ matrix obtained by interchanging the rows and columns of A: the first row of A becomes the first column of the transpose of A and so on.

*Example*:

$$A = \begin{pmatrix} 1 & 2 \\ 4 & 5 \\ 7 & 8 \end{pmatrix},\ A' = \begin{pmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \end{pmatrix}$$

□

Note that the transpose of a matrix $A$ is usually denoted as $A'$ or $A^T$.

The following ***rules*** apply to the transpose:

$$\begin{aligned} (A+B)' &= A' + B' \\ (A-B)' &= A' - B' \\ (A')' &= A \\ (cA)' &= cA' \\ (AB)' &= B'A' \end{aligned}$$

*Example:* Given A,B,C square matrices with the same number of rows, simplify the following expression using matrix algebra rules:

$$((AB)'C + 2A - 3(B'C + \alpha A))\, B' - A(C - \beta I)B'$$

and arrive at

$$(2 - 3\alpha + \beta)AB' - (A + B'A' - 3B')CB'$$

Follow the following steps:

1. Distribute and apply transpose properties:

$$(AB)'C = B'A'C$$

2. Expand and combine like terms:

$$(B'A'C + 2A - 3B'C - 3\alpha A)B' - ACB' + \beta AB'$$

3. Distribute $B'$:

$$B'A'CB' + 2AB' - 3B'CB' - 3\alpha AB' - ACB' + \beta AB'$$

4. Combine like terms:

$$2AB' - ACB' - 3\alpha AB' + \beta AB' + B'A'CB' - 3B'CB'$$

5. Final simplification:

$$(2 - 3\alpha + \beta)AB' - (A + B'A' - 3B')CB'$$

□

## 5.2 The determinant of a matrix

To each square matrix is associated a number called the determinant. The determinant of a $1 \times 1$ matrix is simply the entry of the matrix:

$$det(a) = a$$

Note that the determinant of a matrix A is usually denoted as $|A|$ or $det(A)$. The determinant of a $2 \times 2$ matrix is

$$\text{given } A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, |A| = a_{11}a_{22} - a_{12}a_{21}$$

This is equivalent to write

$$|A| = a_{11}|a_{22}| - a_{12}|a_{21}|$$

where the first term on the RHS is the $(1,1)$th entry of A times the determinant of the submatrix obtained by deleting from A the row and columns which containt that entry; the second term is the $(1,2)$th entry times the determinant of th esubmatrix obtained by deleting from A the row and column which contain the entry. The terms alternate in sign.

The determinant of a $3 \times 3$ matrix is a bit more complicated to get, but it exploits the same logic mentioned in the paragraph above:

$$\text{given } A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}, |A| = a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12}a_{21}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

where the first term on the RHS is the $(1,1)$th entry of A times the determinant of the submatrix obtained by deleting from A the row and columns which containt that entry; the second term is the $(1,2)$th entry times the determinant of the submatrix obtained by deleting from A the row and column which contain the entry;the third term is te $(1,3)$th entry times te determinant of the submatrix obtained y deleting from A the row and column which contain the entry. The terms alternate in sign (or alternatively if $i + j$ of the specific entry is even a plus must be added before the entry, while if it is odd a minus must be added).

The above formula is only one of the possible formula to calculate the determinant of a $3 \times 3$ matrix. The general recipe is

(a) Select a row or a column of the matrix $A_{i \times j}$

(b) Take the first entry of the row/column considered and multiply it by

- the determinant of the submatrix obtained by deleting from A the row and columns which containt that entry (such a determinant is called the $(i,j)$th minor of $A$)
- $+1$ if, given the $(i,j)$th entry, the sum $i + j$ is even or by $-1$ if the sum is odd (such a $+1/-1$ is called the $(i,j)$th cofactor of $A$)

(c) Do the same of all the entries of the row/column

(d) Sum the above quantities

In econometrics we will be often interested in making sure a matrix is **nonsingular**, that is a matrix has a determinant different from zero.

***Properties of the determinant***:

- $Det(\mathbf{0}) = 0$
- $Det(I) = 1$
- $Det(A) = Det(A')$
- If $A$ has one row/column of zeros $\rightarrow Det(A) = 0$
- If $A$ has rows/columns linearly dependent $\rightarrow Det(A) = 0$
- If $A$ is diagonal (or triangular), the determinant is the product of the elements in the diagonal
- If $A$ and $B$ are both $n \times n$, then $det(AB) = Det(A) \cdot Det(B) = Det(BA)$
- If $A$ and $B$ are both $n \times n$, then $det(AB) \neq 0 \Leftrightarrow Det(A) \neq 0 and Det(B) \neq 0$

## 5.3 The inverse of a matrix

The inverse of Matrix is the matrix that on multiplying with the original matrix results in an identity matrix. For any matrix A, its inverse is denoted as $A^{-1}$.

$$AA^{-1} = A^{-1}A = I_n$$

Note that both the matrix and its inverse must be squared of size $n \times n$ and that a matrix is invertible if and only if the matrix is nonsingular. Be also aware that

$$Det(AA^{-1}) = Det(A) \cdot Det(A^{-1}) = Det(I) = 1$$

What is the inverse of a $2 \times 2$ matrix? The inverse of $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is

$$A^{-1} = \frac{1}{det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

where the matrix on the RHS is called the adjoint matrix. Note that indeed if the determinant is equal to zero you cannot get the inverse.

## 5.4 Eigenvalues and eigenvectors

Eigenvalues and Eigenvectors are scalar and vector quantities associated to a square matrix. More specifically, eigenvectors are vectors that are associated with a matrix such that when the matrix multiplies the eigenvector, the resulting vector is a scalar multiple of the eigenvector. The equation to solve to find the determinant are:

$$Av^R = \lambda v^R \text{ or } v^L A = v^L \lambda \tag{4}$$

where $A$ is the square matrix, $v$ is the eigenvector and $\lambda$ is the eigenvalue. Note that the null vector cannot be a eigenvector of a matrix. Why do we have two equation? Because there are two types of eigenvectors, The eigenvector which is multiplied by the given square matrix from the right-hand side is called the right eigenvector, describe by the equation on the right and denoted as $v^R$, and The eigenvector which is multiplied by the given square matrix from the left-hand side is called the left eigenvector, describe by the equation on the left and denoted as $v^L$,

Eigenvector and eigenvalues are of interest to economist because they can be used to study the stability of a time-series process, for example.

How to find eigenvectors and eigenvalues of a matrix $A$? Let's consider for simplicity only the case of the righe eigenvector. We will start by rewrite the above equation as

$$\begin{aligned} &Av = \lambda v \\ &Av - \lambda v = 0 \\ &(A - I\lambda)v = 0 \end{aligned}$$

We then focus on finding the $\lambda s$ and they are found by using the equation $|A - I\lambda| = 0$ (which is called characteristic equation). We will get various values of $\lambda$ as $\lambda_1$ ,$\lambda_2$, ..., $\lambda_n$. Once found the $\lambda s$ one can again exaploit the above equation to find the associated eigenvector $v$.

*Example:* Let's compute the eigenvalues and eigenvectors of $A = \begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix}$

We start by writing the above equation:

$$\begin{aligned} (A - I\lambda)v &= 0 \\ \left( \begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \lambda \right) v &= 0 \end{aligned}$$

We first focus on finding the $\lambda s$ and they are found by using the equation $|A - I\lambda| = 0$:

$$\begin{aligned} \left| \begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \lambda \right| &= 0 \\ \left| \begin{pmatrix} 3-\lambda & 1-0 \\ 2-0 & 2-\lambda \end{pmatrix} \right| &= 0 \\ (3-\lambda)(2-\lambda) - 1 \cdot 2 &= 0 \\ \lambda^2 - 5\lambda + 1 &= 0 \end{aligned}$$

We solve it:

$$\lambda_{1,2} = \frac{5 \pm \sqrt{25-4}}{2} = \frac{5 \pm \sqrt{21}}{2}$$

In this situation we have two eigenvalues, they are $\lambda_1 = \frac{5+\sqrt{21}}{2}$ and $\lambda_2 = \frac{5-\sqrt{21}}{2}$

□

***Properties of the eigenvalues***:

- The eigenvalies of a diagonal (or triangular) matrix are the elements in the diagonal
- if $\lambda$ is eigenvalues of A, then it is also the eigenvalue of $A'$
- if $\lambda$ is eigenvalues of A, then $\alpha \cdot \lambda$ is the aigenvalue of $\alpha\lambda$. The same applied to eigenvectors.

# 6 Lecture 5

Let's start by introducing some useful concepts.

### 6.0.1 Minors of order k vs. minors of element $a_{ij}$

Let B a $m \times n$ matrix. Let $\tilde{B}$ any $k \times k$ *square* submatrix obtained considering $k$ rows and $k$ columns of $B$, then the $det(\tilde{B})$ is called the **minor of $B$ of order $k$**.

So the order of a minor is the number of its rows/columns.

*Example*: Consider the following matrix

$$A = \begin{pmatrix} 1 & -2 & -1 & 0 \\ 3 & -6 & -3 & 1 \end{pmatrix}$$

Possible minors of order 2:

$$Det \begin{pmatrix} 1 & -2 \\ 3 & -6 \end{pmatrix}, Det \begin{pmatrix} -2 & -1 \\ -6 & -3 \end{pmatrix}, Det \begin{pmatrix} -1 & 0 \\ -3 & 1 \end{pmatrix}, Det \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}, Det \begin{pmatrix} 1 & -1 \\ 3 & -3 \end{pmatrix}, Det \begin{pmatrix} -2 & 0 \\ -6 & 1 \end{pmatrix}$$

Note that I have to consider only squared matrix, so given any matrix $A_{m\times n}$, the maximum order of any minor is $k \leq min(m, n)$. □

Let A a $n \times n$ square matrix with $n \geq 2$ and let an element in it be denoted as $a_{ij}$, where $i$ denotes the row and $j$ the column. The minor $M_{ij}$ of $A$ is the determinant of the $(n-1) \times (n-1)$ submatrix formed by removing the $i$-th row and the $j$-th column. To be more specific, this is called **minors of element** $a_{ij}$

Take some time to focus on the difference in the definition of the minors of order k and of the minors of element $a_{ij}$, they are two very similar concepts, but the first applies to any matrix, the second one only to square matrices.

*Example*: Consider the following matrix

$$A = \begin{pmatrix} 1 & 0 & 3 \\ 2 & 3 & 1 \\ 0 & -2 & 2 \end{pmatrix}$$

- $M_{2,1} = Det \begin{pmatrix} 0 & 3 \\ -2 & 2 \end{pmatrix} = 6$
- $M_{3,3} = Det \begin{pmatrix} 1 & 0 \\ 2 & 3 \end{pmatrix} = 3$

□

The product $A_{ij} = (-1)^{i+j} M_{ij}$ is called the **cofactor of the element** $a_{ij}$.

*Example*: Consider the following matrix

$$A = \begin{pmatrix} 1 & 0 & 3 \\ 2 & 3 & 1 \\ 0 & -2 & 2 \end{pmatrix}$$

- $A_{2,1} = (-1)^3 \cdot Det \begin{pmatrix} 0 & 3 \\ -2 & 2 \end{pmatrix} = -6$
- $A_{3,3} = (-1)^6 \cdot Det \begin{pmatrix} 1 & 0 \\ 2 & 3 \end{pmatrix} = 3$

□

#### 6.0.2 Principal Minors and Leading Principal Minors of a Matrix

Let A a $n \times n$ square matrix. The minors got after considering $k$ columns and the *same* $k$ rows from $A$ is called **principal minor** of $A$ of order $k$.

*Example*: Consider the following matrix

$$A = \begin{pmatrix} 1 & 2 & 4 \\ 3 & 1 & 2 \\ 4 & 3 & 6 \end{pmatrix}$$

- The principal minors of order 1: 1,1,6;
- The principal minors of order 2:

$$det \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix} = -5$$
$$det \begin{pmatrix} 1 & 4 \\ 4 & 6 \end{pmatrix} = -10$$
$$det \begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix} = 0$$

- The principal minor of order 3: $det(A) = 0$;

□

Let A a $n \times n$ square matrix. The minor got after considering the first $k$ columns and the first $k$ rows of $A$ is called **leading principal minor** of $A$ of order $k$.

*Example*: Consider the following matrix

$$A = \begin{pmatrix} 1 & 0 & 3 \\ 2 & 3 & 1 \\ 0 & -2 & 2 \end{pmatrix}$$

- The leading principal minor of order 1 is 1;
- The leading principal minor of order 2 is $det \begin{pmatrix} 1 & 0 \\ 2 & 3 \end{pmatrix} = 3$;
- The leading principal minor of order 3 is $det(A) = -4$;

□

## 6.1 Rank

The rank of a matrix A is the maximum number of linearly independent rows (or equivalently of linearly independent columns).

So the rank (r) is a number and is smaller or equal than the number of rows (and of columns).

*Example*:

Consider the following matrix

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

note that here the third column is a linear combination of the first two (because if you sum the first two columns you get the third one). We will see that the rank of this matrix is 2. □

We can also use the minors to find the rank of a matrix: Given $A_{m \times n}$, the rank of $A$ is the maximum order of its non null minors.

*Example*:

$$A = \begin{pmatrix} 1 & 2 & -1 & 4 \\ 3 & 1 & 2 & 2 \\ 4 & 3 & 1 & 6 \end{pmatrix}$$

let's check if there are minors different from zero.

- minors of order 1: yes because at least one element is different from 0
- minors of order 2: yes, for example $det \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix} = -5$
- minors of order 3: no, in fact

$$Det \begin{pmatrix} 1 & 2 & -1 \\ 3 & 1 & 2 \\ 4 & 3 & 1 \end{pmatrix} = 0, Det \begin{pmatrix} 1 & 2 & 4 \\ 3 & 1 & 2 \\ 4 & 3 & 6 \end{pmatrix} = 0, Det \begin{pmatrix} 1 & -1 & 4 \\ 3 & 2 & 2 \\ 4 & 1 & 6 \end{pmatrix} = 0, Det \begin{pmatrix} 2 & -1 & 4 \\ 1 & 2 & 2 \\ 3 & 1 & 6 \end{pmatrix} = 0,$$

The rank of $A$ is 2. □

In the next part of the class, we introduce several concepts essential for understanding the Second Order Condition in optimization. In the case of functions defined on a single variable, the second derivative helps ensure that a critical point is a maximum by indicating the concavity of the function at that point. In multivariate optimization, we analyze the concavity of a function by examining the definiteness of the Hessian matrix.

## 6.2 Gradient vs. Jacobian

The gradient is the vector of first-order partial derivatives of a **scalar-valued function**, that is a function $f$ that returns a scalar value ($f : \mathbb{R}^n \to \mathbb{R}$).

The **Gradient** vector of a function $f()$ at a point $p$ is

$$Gradient = \nabla f(p) = \begin{pmatrix} \frac{\partial f(p)}{\partial x_1} \\ \frac{\partial f(p)}{\partial x_2} \\ \vdots \\ \frac{\partial f(p)}{\partial x_n} \end{pmatrix} \tag{5}$$

Geometrically, the gradient measures the direction of the fastest growth of a function.

The Jacobian is a matrix that holds all first-order partial derivatives of a **vector-valued function**, that is a function $f$ that returns a vector ($f : \mathbb{R}^n \to \mathbb{R}^m$). To understand better, consider the case of $f : \mathbb{R}^2 \to \mathbb{R}^2$

The **Jacobian** matrix vector of a function $f = \begin{pmatrix} h(x,y) \\ g(x,y) \end{pmatrix}$ at a point $(p_x, p_y)$ is

$$Jacobian = J(p_x, p_y) = \begin{pmatrix} h_x(p_x, p_y) & h_y(p_x, p_y) \\ g_x(p_x, p_y) & g_y(p_x, p_y) \end{pmatrix}$$

Note that the Jacobian matrix can also be written as a vector of gradients:

$$Jacobian = Df = J(p_x, p_y) = \begin{pmatrix} h_x & h_y \\ g_x & g_y \end{pmatrix} = \begin{pmatrix} \nabla' h(p) \\ \nabla' g(p) \end{pmatrix}$$

*Example*: Consider the function

$$f(x, y) = \begin{pmatrix} e^x + y \\ x + y^2 \end{pmatrix}$$

The Jacobian matrix is

$$J(x, y) = \begin{pmatrix} e^x & 1 \\ 1 & 2y \end{pmatrix}$$

□

## 6.3 Hessian matrix

The Hessian matrix is the matrix of the second-order partial derivatives of a function **scalar-valued function**, that is a function $f$ that returns a scalar value ($f : \mathbb{R}^n \to \mathbb{R}$).

The Hessian matrix of a function $f()$ at a point $p$ is

$$Hessian = \nabla^2 f(p) = \begin{pmatrix} \frac{\partial^2 f(p)}{\partial x_1^2} & \frac{\partial^2 f(p)}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f(p)}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f(p)}{\partial x_2 \partial x_1} & \frac{\partial^2 f(p)}{\partial x_2^2} & \cdots & \frac{\partial^2 f(p)}{\partial x_2 \partial x_n} \\ \vdots & & \ddots & \vdots \\ \frac{\partial^2 f(p)}{\partial x_n \partial x_1} & \frac{\partial^2 f(p)}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f(p)}{\partial x_2 \partial x_n^2} \end{pmatrix}$$

Note that the Hessian matrix is **symmetric** and that sometimes is also denoted as $D^2 f(p)$.

*Example*: Compute the Hessian of the following function $f(x, y) = x^{\frac{1}{3}} y^{\frac{2}{3}}$

$$\nabla^2 f(x, y) = \begin{pmatrix} \frac{\partial^2 f(x,y)}{\partial x^2} & \frac{\partial^2 f(x,y)}{\partial x \partial y} \\ \frac{\partial^2 f(x,y)}{\partial y \partial x} & \frac{\partial^2 f(x,y)}{\partial y} \end{pmatrix} = \begin{pmatrix} -\frac{2}{9} x^{-\frac{5}{3}} y^{\frac{2}{3}} & \frac{2}{9} x^{-\frac{2}{3}} y^{-\frac{1}{3}} \\ \frac{2}{9} x^{-\frac{2}{3}} y^{-\frac{1}{3}} & -\frac{2}{9} x^{\frac{1}{3}} y^{-\frac{4}{3}} \end{pmatrix}$$

□

## 6.4 Matrix Definiteness

Positive definite and negative definite Hessian matrices are useful concepts in multivariate optimization.

Given a function $f(x)$ and its Hessian matrix $H$, the matrix $H$ is said to be **negative definite** if $xHx \leq 0$ for all $x \neq 0$, where $x$ is the vector of variables ($x \in \mathbb{R}^n$), and if $x'Hx = 0$ only if $x = 0$, that is $x'Hx$ is zero only in zero. Similarly, it is possible to define positive matrices.

In some cases, the condition $xHx = 0$ may occur for some non-zero vectors $x$. In such situations, the matrix is referred to as **semidefinite**:

- $H$ is **positive semidefinite** if $x'Hx \geq 0$ for all $x \in \mathbb{R}^n$, with $x'Hx = 0$ for $x = 0$ and some $x \neq 0$.
- $H$ is **negative semidefinite** if $x'Hx \leq 0$ for all $x \in \mathbb{R}^n$, with $x'Hx = 0$ for $x = 0$ and some $x \neq 0$.

Why do we need these definitions? Because a function $f$ is

- Strictly concave if and only if its Hessian matrix is negative definite
- Strictly convex if and only if its Hessian matrix is positive definite
- Convex if and only if its Hessian matrix is positive semidefinite
- Concave if and only if its Hessian matrix is negative semidefinite

More concretely, how to determine the definiteness of a function?

### 6.4.1 Determine matrix definiteness

- The Hessian Matrix is **negative definite** if and only if all *leading principal minors* of the Hessian are *strictly* positive if their order is even and *strictly* negative if their order is odd (or equivalently all eigenvalues of the Hessian are strictly negative)
- The Hessian Matrix is **positive definite** if and only if all *leading principal minors* of the Hessian are *strictly* positive (or equivalently all eigenvalues of the Hessian are strictly positive)
- The Hessian Matrix is **negative semidefinite** if and only if all the *principal minors* of the Hessian are *nonnegative* ($\geq 0$) if their order is even and *nonpositive* ($\leq 0$) if their order is odd (or equivalently all eigenvalues of the Hessian are nonpositive and exist at least one eigenvalue which is equal to zero)
- The Hessian Matrix is **positive semidefinite** if and only if all the *principal minors* of the Hessian are *non negative* ($\geq 0$) (or equivalently all eigenvalues of the Hessian are nonnegative and exist at least one eigenvalue which is equal to zero)

*Example*: Let's check the definiteness of the following function

$$-x - 2y^2$$

We compute the Hessian

$$\begin{pmatrix} 0 & 0 \\ 0 & -4 \end{pmatrix}$$

- We compute the leading principal minors: the leading principal minor of order 1 is 0, the leading principal minor of order 2 is 0. The matrix is not negative definite or positive definite.
- Let's now compute the other principal minors, we computed all the principal minors of order 2 (equal to 0), and only the principal minors of order 1 are left. They are all 0 except for one which is $-4$, given that all the even principal minors of the Hessian are nonnegative and all the odd are nonpositive, the Hessian matrix is negative semidefinite.

Given that the hessian is negative semidefinite the function is concave. Note that we could also have equivalently calculated the eigenvalues to determine the definitiveness of a matrix. □

#### 6.4.2 Summary

We summarize the above information in the following conditions:

| | | | | | | |
|---|---|---|---|---|---|---|
| Negative Hessian matrix | $\Longleftrightarrow$ | All leading principal minors of the Hessian are strictly positive if their order is even and strictly negative if their order is odd | or | All eigenvalues of the Hessian are strictly negative | $\Longleftrightarrow$ | Function is strictly concave |
| Positive Definite Hessian matrix | $\Longleftrightarrow$ | All leading principal minors of the Hessian are strictly positive | or | All eigenvalues of the Hessian are strictly positive | $\Longleftrightarrow$ | Function is strictly convex |
| Negative Semidefinite Hessian matrix | $\Longleftrightarrow$ | All principal minors of the Hessian are nonnegative if their order is even and nonpositive if their order is odd | or | All eigenvalues of the Hessian are nonpositive and exist at least one eigenvalue which is equal to zero | $\Longleftrightarrow$ | Function is concave |
| Positive Semidefinite Hessian matrix | $\Longleftrightarrow$ | All principal minors of the Hessian are nonnegative | or | All eigenvalues of the Hessian are nonnegative and exist at least one eigenvalue which is equal to zero | $\Longleftrightarrow$ | Function is convex |

#### 6.4.3 Special case: diagonal matrix

Given that the determinant of a diagonal matrix is the product of its diagonal elements, we get that for a diagonal matrix:

- The Hessian Matrix is **negative definite** if and only if all the diagonal elements are negative;
- The Hessian Matrix is **positive definite** if and only if all the diagonal elements are positive;
- The Hessian Matrix is **negative semidefinite** if and only if all the diagonal elements are nonpositive;
- The Hessian Matrix is **positive semidefinite** if and only if all the diagonal elements are nonnegative

## 6.5 FINAL EXERCISE

Consider the following function: $f(x) = -x^2 - y^2 + 4x + 6y$

(a) Find the maximum of the function subject to the following constraint $0 \geq +2x + y - 3$.

(b) From now onwards forget about the constraint. Find an approximation of the function around the point $(x_0, y_0) = (1, 1)$

(c) Consider the following equation: $f(x) = 0$, find how the value of $y$ changes when you change $x$ at $(0, 0)$

(d) Find the inverse of the Hessian.

(a) Let's solve

$$\begin{aligned} \text{maximize} \quad & -x^2 - y^2 + 4x + 6y \\ \text{subject to} \quad & 0 \geq +2x + y - 3 \end{aligned}$$

The first thing to do is to rewrite the constraint as

$$-2x - y + 3 \geq 0$$

Now we follow the step above.

**Assumptions' check**

i. The domain is $\mathbb{R}^2$ which is open and convex

ii. $-x^2 - y^2 + 4x + 6y$ is continuous since it is the sum of two continuous functions. To check if differentiable, we check if its partial derivatives exist and are continuous. We compute the gradient:

$$\frac{\partial f}{\partial x} = -2x + 4$$
$$\frac{\partial f}{\partial x} = -2y + 6$$

which are both continuous.

iii. Check if the function $-x^2 - y^2 + 4x + 6y$ is concave or strictly concave. We compute the Hessian:

$$\nabla^2 f(x, y) = \begin{pmatrix} \frac{\partial^2 f(x,y)}{\partial x^2} & \frac{\partial^2 f(x,y)}{\partial x \partial y} \\ \frac{\partial^2 f(x,y)}{\partial y \partial x} & \frac{\partial^2 f(x,y)}{\partial y} \end{pmatrix} = \begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix}$$

- The leading principal minor of order 1 is -2 and of order 2 is +4. The matrix is neither positive or negative definite.
- The remaining principal minor is -2. All even principal minors are nonnegative and all positive principal minors are nonpositive, so the Hessian is Negative Semidefinite.

Since the Hessian is negative Semidefinite the function is concave in the domain.

iv. $-2x - y + 3$ is linear so ok.

v. $-2x - y + 3$ is linear so ok.

vi. The point $(0, 0)$ belongs to the domain and satisfies the constraint. So ok.

**Theorem Application**

i. Construct the lagrangian

$$L(x, y, \lambda, \mu) = -x^2 - y^2 + 4x + 6y + \lambda(-2x - y + 3)$$

ii. K-T conditions:

$$\begin{aligned}
\frac{\partial L}{\partial x}(x^*, y^*, \lambda^*, \mu^*) &= -2x + 4 - 2\lambda = 0 \\
\frac{\partial L}{\partial y}(x^*, y^*, \lambda^*, \mu^*) &= -2y + 6 - \lambda = 0 \\
\lambda^*(g(x^*, y^*)) = \lambda(-2x - y + 3) &= 0 \\
\lambda^* &\geq 0 \\
g(x^*, y^*) = -2x - y + 3 &\geq 0
\end{aligned}$$

iii.
- Assume $\lambda = 0$, we get $2x + 4 = 0$ and $-2y + 6 = 0$, so $x = 2$ and $y = 3$, but the K-T condition on the constraint is not satisfied
- Assume $\lambda\ 0$. We get $-2x - y + 3 = 0$, $-2x + 4 - 2\lambda = 0$ and $-2y + 6 - \lambda = 0$, so that from the first we have $y = -2x + 3$, which we plug in the third one and get $4x - 6 + 6 - \lambda = 0$ and $\lambda = 4x$, we plug it in the second one and get $-2x + 4 - 8x = 0$, and finally we have $x = 2/5, y = 11/5, \lambda = 8/5$, which satisfies all the conditions.

The point

$$(\frac{2}{5}, \frac{11}{5})$$

is the solution of the maximization problem.

iv. We apply the total differential:

$$\begin{aligned}
df &= \nabla f(1, 1) \cdot dx \\
&= (-2x + 4)dx + (-2y + 6)dy \\
&= (-2 \cdot 1 + 4)dx + (-2 \cdot 1 + 6)dy \\
&= (-2 + 4)dx + (-2 + 6)dy \\
&= 2dx + 4dy
\end{aligned}$$

this is a linear approximation of the function around the point $(1, 1)$. For example, suppose we would be interested in how much the value of the function changes when we move from $(1, 1)$ to $(1, 2)$, therefore when $dx = (1 - 1) = 0$ and $dy = (2 - 1) = 1$, we would get

$$df = 2 \cdot 0 + 4 \cdot 1 = 4$$

ok this was an approximation, let's check its value by computing

$$\begin{aligned}
f(1, 1) &= -x^2 - y^2 + 4x + 6y = -1 - 1 + 4 + 6 = 8 \\
f(1, 2) &= -x^2 - y^2 + 4x + 6y = -1 - 4 + 4 + 12 = 11
\end{aligned}$$

and we get that

$$f(1,2) - f(1,1) = 3$$

The value of the two values, 3 and 4, is very close, obviously the total differential is a linear approximation so the two values differs a bit.

v. We apply the IFT recipe

A. We write down the function and compute the derivative

$$\begin{aligned} f(x,y) &= -x^2 - y^2 + 4x + 6y \\ \frac{\partial f(x,y)}{\partial y} &= -2y + 6 \\ \frac{\partial f(x,y)}{\partial x} &= -2x + 4 \end{aligned}$$

B. $x$ is the exogenous variable and $y$ the endogenous one

C. We check the assumptions

- The function and the derivatives are continuous in the domain
- We select as points $(0,0)$
  - $(0,0)$: $G(0,0) = 0$ and $\frac{\partial G(1,0)}{\partial y} = 6$, therefore the IFT can be applied at $(0,0)$ by solving for $y$.

D. We apply the IFT at $(0,0)$ and compute the derivative

$$F'(x) = -\frac{\frac{\partial G}{\partial x}(0,0)}{\frac{\partial G}{\partial y}(0,0)} = -\frac{-2 \cdot 0 + 4}{-2 \cdot 0 + 6} = -\frac{4}{6}$$

vi. The Hessian is

$$\begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix}$$

to find the inverse we need the determinant that is $+4$. We apply the formula

$$\begin{aligned} \nabla^2 f(x,y)^{-1} &= \frac{1}{det(\nabla^2 f(x,y))} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \\ &= \frac{1}{4} \begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix} \\ &= \begin{pmatrix} -\frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} \end{aligned}$$

# A K-T Exercise seen in class

In class we saw the following exercise.

$$\begin{aligned} \text{maximize} \quad & -x^2 - y \\ \text{subject to} \quad & x + y - 1 \geq 0 \\ & x - 2y^2 \geq 0 \end{aligned}$$

So we have the same objective function but two constraints now. We haven't seen this in the theorem, but we can simply verify that the assumptions in the new constraint holds too and generalize the KKT conditions.

The steps on the assumption related to the objective function are the same as above

**Assumptions' check**

i. We are in $\mathbb{R}^2$ which is open and convex

ii. The function is the sum of two continuous functions

iii. The gradient is

$$\nabla f(x) = \begin{pmatrix} -2x \\ -1 \end{pmatrix}$$

since both the components are continuous, the gradient is differentiable.

iv. $x + y - 1$ is linear

v. $x + y - 1$ is linear

vi. $x - 2y^2$ is the sum of continuous functions. To check the differentiability we check the gradient.

$$\frac{\partial g_2}{\partial x} = 1$$
$$\frac{\partial g_2}{\partial y} = -4y$$

The gradient is continuous, so the function is differentiable.

vii. $x - 2y^2$ is concave. So ok.

viii. The point $(3, 1)$ belongs to the domain and satisfies the constraint. So ok.

**Theorem Application**

i. Construct the lagrangian

$$L(x, y, \lambda, \mu) = -x^2 - y + \lambda(x + y - 1) + \mu(x - 2y^2)$$

ii. K-T conditions:

$$\begin{aligned}
\frac{\partial L}{\partial x}(x^*, y^*, \lambda^*, \mu^*) = -2x + \lambda + \mu = 0\\
\frac{\partial L}{\partial y}(x^*, y^*, \lambda^*, \mu^*) = -1 + \lambda - 4\mu y = 0\\
\lambda(x + y - 1) = 0\\
\mu(x - 2y^2) = 0\\
\lambda \geq 0\\
\mu \geq 0\\
x + y - 1 \geq 0\\
x - 2y^2 \geq 0
\end{aligned}$$

iii.
- Assume $\lambda = 0$, we have two cases: $\mu = 0$ or $\mu > 0$.
  If $\mu = 0$, the second condition is $-1 = 0$, so not verified.
  If $\mu > 0$, we get the following conditions

  $$\begin{aligned}
  x = 2y^2\\
  -2x + \mu = 0\\
  -1 - 4\mu y = 0\\
  \lambda = 0\\
  \mu > 0\\
  x + y - 1 \geq 0
  \end{aligned}$$

  by combining the first with the second we get $\mu = 4y^2$. We then combine it with the third and get $-1 - 16y^3 = 0$, that is $y = (-\frac{1}{16})^{\frac{1}{3}}$ and therefore $\mu = 4(-\frac{1}{16})^{\frac{2}{3}}$ and $x = 2(-\frac{1}{16})^{\frac{2}{3}}$. SUing the calculator $\mu \approx 0.62$ and $x \approx 0.31$ and $y \approx 0.16$, therefore the last constraint is not satisfied
- Assume $\lambda > 0$, we have two cases: $\mu = 0$ or $\mu > 0$.
  If $\mu = 0$, we get the following conditions

  $$\begin{aligned}
  -2x + \lambda = 0\\
  -1 + \lambda = 0\\
  x + y - 1 = 0\\
  \lambda > 0\\
  \mu = 0\\
  x - 2y^2 \geq 0
  \end{aligned}$$

  From the second condition we get $\lambda = 1$, we plug it in the first condition and get $x = \frac{1}{2}$. From the third condition we get $y = \frac{1}{2}$. We now check if the last condition is satisfied: $\frac{1}{2} - 2\frac{1}{4} = \frac{1}{2} - \frac{1}{2} = 0$. So the point $(\frac{1}{2}, \frac{1}{2})$ satisfies the system.

If $\mu > 0$, we get the following conditions

$$\begin{aligned}
-2x + \lambda + \mu &= 0 \\
-1 + \lambda - 4\mu y &= 0 \\
x + y - 1 &= 0 \\
x - 2y^2 &= 0 \\
\lambda &> 0 \\
\mu &> 0
\end{aligned}$$

Combining the third and the fourth one we get $2y^2 + y - 1 = 0$, the results are $y_1 = \frac{1}{2}$ and $y_2 = -1$. For $y_1 = \frac{1}{2}$, we get $x = \frac{1}{2}$ wich has already been found above. For $y = -1$, we get $x = 2$ and the conditions left are

$$\begin{aligned}
-4 + \lambda + \mu &= 0 \\
-1 + \lambda + 4\mu &= 0 \\
\lambda &> 0 \\
\mu &> 0
\end{aligned}$$

from which we get $\mu = -1$ so the system is not satisfied. The points

$$(\frac{1}{2}, \frac{1}{2})$$

is the solution of the maximization problem.